Seized Knowledge: Wartime Vesting and the Long-Run Growth of U.S. Innovation

Jaysa Rafi[†] Lovepreet Singh[*]

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Abstract

During World War II, the United States seized U.S. patents held by residents of the Axis and Axis-occupied countries. Beginning in 1942, the Alien Property Custodian seized about 38,000 patents and licensed them to any American applicant. We digitize the seized patents and construct a measure of each patent subclass's exposure to the seizure. We find that in exposed subclasses, U.S.-origin patenting rose sharply in 1942, to about 27 percent above unexposed subclasses, and remained 38 percent higher as late as 1975. The increase is larger and more persistent for patents assigned to U.S. firms, and it is not explained by wartime research funding. Patenting in the United States by inventors from the Axis and occupied countries fell by about 78 percent in the exposed subclasses and had not recovered by 1975. The seizure persistently shifted invention in the exposed technologies from foreign to American inventors.

[*]We thank Richard Akresh, David Albouy, Alex Bartik, and Dan Bernhardt for their valuable feedback.
[†]Visiting Assistant Professor, Williams College. Email: jr37@williams.edu.
[*]Ph.D. Candidate, Department of Economics, University of Illinois Urbana-Champaign. Email: ls12@illinois.edu.

1 Introduction

Industrial policy is any government policy that targets the structure of economic activity in pursuit of a public goal, and its instruments run well beyond subsidies and protection to the public provision of the inputs that firms lack. Foreign technology is one such input. A state can pay to develop it at home, obtain it from its foreign owners in exchange for market access, or force its transfer by seizing their intellectual property. Much less is known about the third of these, forced technology transfer, and in particular about a transfer that is permanent, covers a wide array of technology areas, and is complemented by other strategic state actions against the foreign owners.[1]

In this paper we study one such transfer, the seizure of foreign-owned patents in the United States during World War II. Under the Trading with the Enemy Act, the Office of the Alien Property Custodian took title to the U.S. patents held by residents of the Axis countries and of the countries they occupied. By June 1943 it held 36,675 patents, about 5 percent of all unexpired U.S. patents (Office of the Alien Property Custodian, 1943b). The largest holdings belonged to firms, among them IG Farben in chemicals, Telefunken and Siemens in radio and telephony, Bosch and Daimler-Benz in engines, and Zeiss in optics. The Custodian licensed the patents to any American applicant, without exclusivity and, for Axis-owned patents, without royalty for the life of the patent. The owners received no compensation, and Congress later barred the return of German- and Japanese-owned property (Office of Alien Property, Department of Justice, 1949). The United States also collected know-how from German plants (Gimbel, 1990), and the Allies broke up IG Farben (Pöge, 2022). We ask whether this transfer raised U.S. patenting in the technologies in which the vested patents were concentrated. We find that it did. U.S.-origin patenting rose in exposed technologies in the year of the seizure and was 38 percent higher three decades later, while U.S. patenting by inventors located in Axis and occupied countries fell by about 78 percent and did not return.

We assemble a record of 38,294 patents vested by the Custodian, digitized from the five editions of its *Catalog of Vested Patents* published between 1942 and 1944, which list each patent by number under its Patent Office class. We link these records to historical U.S. patent data (Gross and Sampat, 2023).[2] Our design compares patenting before and after 1942 across technology subclasses that the seizure reached to different degrees. In the baseline we call a subclass exposed if the Custodian vested at least one of its pre-war

[1] The closest evidence comes from the compulsory licensing of German-owned chemical patents in the United States after World War I (Moser and Voena, 2012). We discuss this episode further below.

[2] The patent data give every patent filed from 1930 to 1975 its technology subclass, its assignee, and the country of its inventor.

patents, and in robustness checks we use the share of its pre-war patents that the Custodian vested. Because the Custodian vested every unexpired patent held by residents of the Axis and occupied countries, exposure was set by the pre-war patenting of foreign inventors and not by a policymaker's choice of sectors (Office of the Alien Property Custodian, 1943a, pp. 7–8). We apply the same comparison to U.S. patenting by inventors located abroad.

We estimate the difference in patenting between exposed and unexposed subclasses in each year from 1930 to 1975, relative to 1938. For U.S.-origin patents,[3] the estimated differences are close to zero from 1930 to 1941, turn positive in 1942, the year in which the Custodian vested most of the patents, and remain positive through 1975. Relative to 1930 to 1938, exposed subclasses record 27 percent more U.S.-origin patents than unexposed subclasses in 1942 to 1945 and 38 percent more in 1960 to 1975. The increase is more stable for patents assigned to firms. The estimate for all U.S.-origin patents declines to 16 percent in 1946 to 1948, whereas the estimate for patents assigned to firms remains close to 30 percent in each period from 1942 to 1959. For U.S. patents by inventors located in Axis and occupied countries, the estimated differences turn negative in 1939, at the outbreak of war in Europe.[4] From 1947 to 1975, exposed subclasses record about 78 percent fewer of these patents than unexposed subclasses, relative to 1938, and we find no comparable decline for inventors located in Allied and other foreign countries. The foreign owners, unlike their predecessors after World War I, did not come back.

The increase appears in each of the three technology sectors that contain most of the vested patents. Mechanical, chemical, and electrical technologies together account for more than two-thirds of the vested patents.[5] Estimated separately within each of these sectors, the difference in U.S.-origin patents assigned to firms between exposed and unexposed subclasses turns positive in 1942 or 1943 and remains positive through 1975. In chemical technologies, which account for 26 percent of pre-war patents by inventors located in Axis and occupied countries and 12 percent of pre-war U.S.-origin patents, exposed subclasses record about 54 percent more such patents than unexposed subclasses in 1960 to 1975, relative to 1938. In computers and communications, the estimates decline after 1946 and are close to zero after 1965. The response was not confined to the industry in which the foreign firms were strongest.

The estimates are similar under alternative measures of exposure and of patenting. They remain positive in every period from 1942 to 1975 when we measure exposure as

[3]We classify a patent as U.S.-origin when its first-listed inventor is located in the United States.

[4]The coefficients for 1942 to 1946 cannot be estimated, because exposed subclasses record too few such patents in those years.

[5]We use the six technology sectors of Hall et al. (2001). Their residual sector, which groups the technologies outside the other five, accounts for 24 percent of the vested patents, and its estimates follow the same pattern.

the share of a subclass's pre-war patents that the Custodian vested, when we weight patents by the citations they receive, and when we estimate by least squares on the inverse hyperbolic sine of patent counts. The same holds when we include fixed effects for two-digit technology category by year, and when we replace subclasses with CPC main groups. The estimates also change little when we control for the share of a subclass's wartime patents funded by the Office of Scientific Research and Development (Gross and Sampat, 2023). The increase in U.S.-origin patenting in exposed subclasses is therefore not explained by wartime research funding.

We contribute to the empirical literature on industrial policy (Juhász et al., 2024), where we study an unconventional instrument, the seizure of foreign-owned patents. Recent work estimates the effects of temporary trade protection (Juhász, 2018) and of targeted sectoral promotion (Lane, 2025) on industrial development.[6] Both instruments raise the return to production in the targeted industries. The seizure instead transferred the rights to an existing stock of foreign technology. The closest evidence on such a transfer comes from the compulsory licensing of German-owned chemical patents in the United States after World War I, which raised patenting by U.S. inventors in the licensed fields. That program covered a single industry, and German patentees in chemicals resumed patenting in the United States within a decade (Moser and Voena, 2012). The seizure after World War II was permanent and covered many fields of technology.[7] Exposure to it was not chosen by a policymaker, and the patent record extends three decades beyond it, which separates the wartime response from the long-run response.

We also contribute to the literature on international technology transfer. Existing evidence concerns transfers arranged between particular owners and particular recipients, with the owner's agreement. It comes from joint ventures formed as a condition of market access (Holmes et al., 2015; Jiang et al., 2024; Bai et al., 2025) and from programs that supplied machinery and training to chosen firms and plants (Giorcelli, 2019; Giorcelli and Li, 2026).[8] The seizure was a forced transfer. It took the entire stock of U.S. patents held by residents of the Axis and occupied countries, without the owners' consent, and made it available to any American applicant rather than to chosen recipients.

[6] Other work in this literature studies public research contracts (Gross and Sampat, 2023) and the public provision of technical knowledge (Juhász et al., 2026).

[7] After World War I the Custodian sold the patents, and most seized property was later returned to its owners. After World War II the Custodian kept title, licensed Axis-owned patents without royalty, and Congress barred the return of German- and Japanese-owned property (Office of the Alien Property Custodian, 1943a; Office of Alien Property Custodian, 1945; Office of Alien Property, Department of Justice, 1949).

[8] Bai et al. (2025) find that knowledge passed from joint ventures to affiliated Chinese automakers through worker flows and suppliers. Giorcelli and Li (2026) find that the gains from Soviet machinery in Chinese steel plants faded where the machinery came without training.

Finally, we contribute to the literature on World War II and the U.S. innovation system. This literature finds that the wartime research effort had lasting effects on the location and direction of U.S. invention (Gross and Sampat, 2023, 2025; Gross and Roche, 2024). Wartime policy toward patents has received less attention. Gross (2023) finds that secrecy orders, which withheld patent rights and barred disclosure, impeded follow-on invention. We study a wartime patent policy of the opposite kind. The Custodian opened foreign-owned patents to any American applicant and published their contents. We estimate its effects in the same patent data and over the same years, and we control for research contracts in the same technologies.

The paper proceeds as follows. Section 2 describes the historical background. Section 3 describes the data, and Section 4 documents the composition of the vested patents and the variation in exposure across technologies. Section 5 sets out the empirical strategy. Section 6 reports the main results, and Section 7 reports them by technology sector. Section 8 concludes.

2 Historical Background

Before World War II, German firms led in several industries, among them chemicals, electrical equipment, and optics. They patented their inventions in the United States as well as in Germany, and American firms obtained access to these inventions through license agreements. The patents were concentrated in chemical and electrical technologies. These two sectors account for 42 percent of the U.S. patents that inventors located in the Axis and occupied countries filed before the war, compared with 23 percent of the patents of inventors located in the United States. The license agreements often did more than grant a license. Many divided markets between the parties, limited the output and exports of the American licensee, and restricted its research, and in the Custodian's assessment they had made American industry dependent on German supplies and discouraged independent American research (Office of the Alien Property Custodian, 1943b).

The war severed these ties. Trade with Germany ceased, the German owners could no longer enforce their agreements, and the United States needed the same technologies for its own war production. The federal government responded in two ways. It financed new research directly, through the Office of Scientific Research and Development, which awarded more than 2,500 contracts to firms and universities between 1940 and 1945, and in 1942 it seized the U.S. patents of German and other foreign owners and licensed them to American firms. The research contracts paid for inventions that did not yet exist, whereas

the seizure transferred inventions that already existed. After Germany's surrender, the government sent investigators to German plants and laboratories to collect the technical information that the patents did not disclose (Stewart, 1948; Gimbel, 1990).

2.1 The Seizure and Licensing of Foreign-Owned Patents

The seizure was carried out under the Trading with the Enemy Act of 1917. The First War Powers Act of December 1941 amended the Act to allow the President to vest property in the United States in which a foreign country or its nationals held an interest. To vest a property was to take title to it. Executive Order 9095 of March 11, 1942 created the Office of Alien Property Custodian and delegated this power to it. The Custodian vested only patents owned by residents of the Axis countries and of the countries they occupied. It listed from the records of the Patent Office the unexpired patents issued to such residents, excluded those in which Americans held an interest, and vested the rest. The Custodian reported that by December 31, 1942 it had vested every unexpired patent and patent application owned by residents of these countries. By June 30, 1943 it had vested 36,675 patents and 4,757 pending applications. These were about 5 percent of all unexpired U.S. patents, and they fell in all but four of the roughly 300 classes of the Patent Office. The Custodian published its holdings in a *Catalog of Vested Patents*, and it published abstracts of the vested patents. The catalog lists each patent by number under its Patent Office class. Its purpose was to let American manufacturers find the patents and put them to use in war production and post-war research.

The Custodian did not sell the vested patents. It kept title to them and issued licenses. Any American firm or individual could apply. The licenses were nonexclusive and could not be assigned. For patents that had belonged to residents of the Axis countries, the licenses were royalty-free for the life of the patent. The fee was $50 for the first patent and $5 for each further related patent until August 1943, and $15 per patent thereafter. The Custodian gave two reasons for this policy. Keeping title prevented the patents from reverting to their former owners, and licensing without exclusivity prevented any one firm from using them to build a monopoly.[9]

American firms took up the licenses in large numbers. By June 1944 the Custodian had issued 861 licenses to 476 firms and individuals, covering 5,853 patents and applications. By June 1945 it had licensed 7,343 of the 22,000 patents that were available for licensing, and by June 1946 it had issued 2,012 licenses to 1,097 licensees. In 1945, 44 percent of

[9]Patents of residents of the occupied countries were treated differently. They were licensed only against royalties, the Custodian had licensed 38 of them by June 1946, and from 1946 it returned them to their owners.

licensees reported that they had used the patents in production or research. The Custodian did not take every patent of German origin. Where a German owner had assigned a patent outright to an American firm before the war, the Custodian ordinarily vested only the owner's interest in the contract, so the patent itself does not appear in the catalog (Office of Alien Property Custodian, 1945, 1946, 1947).

2.2 World War I and the Years After 1945

The United States had seized German patents once before, and the two episodes ended differently. After World War I the Custodian sold the property it had seized. It sold the American subsidiary of Bayer at auction in 1918, and in 1919 it sold about 4,500 German chemical patents to the Chemical Foundation, a private corporation that licensed them to American firms for royalties. Most of the proceeds were later returned to the former owners, and German firms returned to the American market. U.S. patents by German inventors in chemicals rose from 32 in 1919 to 353 in 1930. After World War II the Custodian kept title to the patents. Nationals of Germany and Japan could again obtain U.S. patents from August 1947, but the War Claims Act of 1948 provided that their seized property would not be returned and that no compensation would be paid (Steen, 2014; Moser and Voena, 2012; Office of Alien Property, Department of Justice, 1949).

The seizure was one part of a wider American program toward German industry. In September 1944 President Roosevelt asked Leo Crowley, who had been the Alien Property Custodian and now headed the Foreign Economic Administration, to study what should be done after Germany's surrender to control its capacity to make war. The Foreign Economic Administration organized Technical Industrial Disarmament Committees, staffed by about 250 experts, which prepared 32 studies of the post-surrender treatment of German industries. The studies covered many of the industries in which the vested patents were concentrated, among them chemicals, electronics, optical instruments, and machine tools, and they also covered German participation in international cartels. Draft reports went to the American military government in Berlin in August 1945. The Potsdam Agreement of the same month provided that German production of metals, chemicals, and machinery would be controlled and restricted, and that the German economy would be decentralized to eliminate cartels, trusts, and other concentrations of economic power. The Allies later broke up IG Farben, and by 1952 its business had passed to three successors, BASF, Bayer, and Hoechst (Foreign Economic Administration, Enemy Branch, 1945; Berlin Conference, 1945; Pöge, 2022).

The patents alone did not always transfer the technology. In 1945 the Custodian

reported that most patents did not disclose enough know-how to make the inventions workable, and that some licensees could not even prepare working drawings from them. It arranged for the Technical Industrial Intelligence Committee to obtain the missing information in Germany. The committee was chaired by Howland Sargeant, who in 1943 had headed the Custodian's Division of Patent Administration, and many of its investigators came from American firms. After Germany's surrender, the Army's Field Information Agency, Technical, supported such investigations until June 1947, and the Department of Commerce sold the resulting reports to the public. The Bosch condenser machine illustrates the process. The Custodian had vested Bosch's U.S. patent on metallized paper condensers and offered licenses during the war, but the patent alone was not enough for a manufacturer to adopt the process. After the war an engineer from Western Electric studied the machine at Bosch's plant in Stuttgart, the machine was shipped to the United States and shown to engineers from 131 firms and agencies, and Western Electric built five similar machines (Office of the Alien Property Custodian, 1943b; Office of Alien Property Custodian, 1946; Electronics, 1946; Gimbel, 1990).

3 Data

Our analysis combines two sources. The first is the Custodian's *Catalog of Vested Patents*, which identifies the patents that the Custodian vested and which we digitize from its five editions. The second is the historical U.S. patent data of Gross and Sampat (2023), which give each patent its filing date, technology class and subclass, assignees, and the location of its first inventor. We link the two sources by patent number and match 38,294 vested patents. From the linked data we build a panel of patent counts by technology subclass and filing year from 1930 to 1975. We measure each subclass's exposure to the seizure by the share of its patents filed before April 4, 1942 that the Custodian vested.

3.1 Alien Property Custodian Vested Patents

We identify the vested patents from the Custodian's *Catalog of Vested Patents*, which lists every patent and pending application that the Custodian vested, by number, under its Patent Office class. We digitized all five editions of the catalog, published between October 1942 and January 1944.[10] The editions contain 45,054 records, 41,827 for granted patents

[10]The editions are the first patent edition (October 2, 1942), the patent application edition (November 10, 1942), the first patent supplement (January 18, 1943), the second patent supplement (October 20, 1943), and the correction supplement (January 13, 1944).

and 3,227 for pending applications. We link the records to the historical patent data by patent number, so we use the granted patents only. After we remove repeated listings and 89 malformed numbers, 40,082 distinct patents remain, and 38,294 of them, or 95.5 percent, match. These 38,294 patents form our sample of vested patents (Table 1), and Figure 1 reproduces the catalog entries for four of them. The catalog accounts for nearly all of the patents that the Custodian vested. Its 40,082 distinct patents compare with 40,616 patents that the Custodian reported having vested by June 30, 1948, and part of the difference reflects patents vested after the last edition of the catalog.

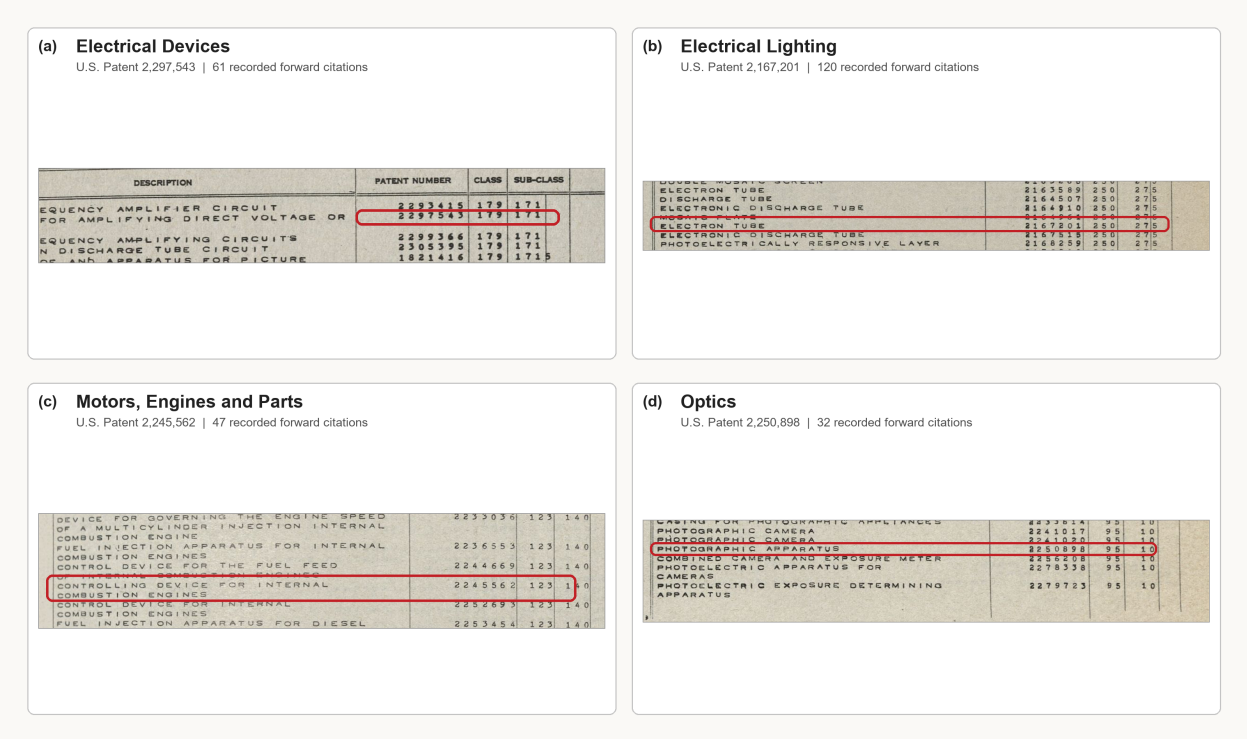


Figure 1: Highly cited vested patents in four technology sectors. Each panel shows the catalog entry for the vested patent with the largest recorded forward-citation count in the indicated NBER two-digit subcategory: Electrical Devices, Electrical Lighting, Motors, Engines and Parts, and Optics. Red outlines identify the selected entries. The catalog reports contemporaneous USPTO classes and subclasses, which may differ from the later classifications used to construct the NBER categories.

3.2 U.S. Patent Data

We draw the outcomes from the historical U.S. patent data of Gross and Sampat (2023). These data start from the USPTO master file of U.S. utility patents, which gives each patent's number, grant date, and USPC class and subclass. Gross and Sampat add application serial numbers, filing dates, front-page citations, assignee names and types, and inventor

locations. We keep the patents filed from 1930 to 1975 that have a USPC class and subclass. We date each patent by its filing year rather than its grant year, because the filing date is closer to the date of the invention. Our outcomes are annual counts of patents in each USPC class and subclass. We count patents separately by the type of their assignee and by the location of their inventor.

Gross and Sampat classify each assignee as a firm, an individual, a government entity, or a university, and a patent may have more than one assignee. We classify a patent as assigned to a firm if at least one of its assignees is a firm. Counts of all patents include every patent, and counts of firm patents include only the patents assigned to a firm. We classify each patent by the location of its first-listed inventor. A patent is U.S.-origin if its first inventor is located in the United States. We group the remaining patents into those whose first inventor is located in the Axis countries, in the occupied countries, or in Allied and other foreign countries. Table 2 lists the countries in each group. Patents without a recorded inventor location enter the counts of all patents but not the counts by origin. For robustness, we also weight each patent by the forward citations it receives within 20 years of its grant. Gross and Sampat obtain the citations from the front pages of later patents, which list the patents they cite only from 1947, so citation weights are low for patents filed before the war.

Our unit of technology is the USPC class and subclass. Subclasses are narrow. Class 526, synthetic resins, for example, contains subclass 526/340, polymers interpolymerized with an aromatic hydrocarbon, which has seven patents filed before the seizure, three of them from IG Farben that the Custodian vested. For descriptive analysis and for estimates by technology, we assign each subclass to the two-digit NBER category of Hall et al. (2001) most common among its patents, and we group these categories into the six NBER sectors. We also build a panel of Cooperative Patent Classification (CPC) main groups from the USPTO's master classification file of July 2026. We use the inventive CPC assignments and aggregate subgroups to main groups. A patent may belong to several main groups, so it can enter more than one cell, but it enters each main group only once. The CPC panel covers filing years 1930 to 1975, and we use it in robustness checks.

3.3 Analysis Sample and Main Variables

Let j index USPC subclasses and t index filing years. Our panel contains every subclass in every year from 1930 to 1975, with a count of zero in years without a patent. For each subclass and year we count all patents, patents assigned to firms, U.S.-origin patents, and U.S.-origin patents assigned to firms. We also count patents whose first inventor is located

in the Axis and occupied countries and patents whose first inventor is located in Allied and other foreign countries. We measure each subclass's exposure to the seizure with the patents filed before April 4, 1942. Let V_j be the number of vested patents in subclass j filed before that date, and let N_j^0 be the number of all patents in the subclass filed before that date. Exposure is the vested share,

$$\text{APC Exposure}_j = \frac{V_j}{N_j^0}. \tag{1}$$

Our baseline uses an indicator for any exposure,

$$\text{Any APC Exposure}_j = \mathbf{1}\{V_j > 0\}, \tag{2}$$

which compares subclasses with at least one vested patent to subclasses with none. We use the vested share, standardized within the estimation sample, as an alternative.

The baseline sample includes the subclasses with at least seven patents filed before April 4, 1942. Seven is the median number, and in smaller subclasses a single vested patent moves the vested share by more than two standard deviations. We control for wartime research funding with a measure built from the data of Gross and Sampat (2023). For each subclass, the OSRD rate is the share of its patents filed from 1941 to 1948 that were funded by the Office of Scientific Research and Development. In robustness checks we interact this rate with filing-year indicators.

4 Descriptive Evidence on APC Patent Exposure

This section documents three features of APC vesting that motivate the empirical design. First, vesting was large and overwhelmingly concentrated among patents originating in Axis and occupied countries. Second, because foreign firms specialized in different technologies, vested patents were distributed unevenly across U.S. patent classes, generating substantial cross-technology variation in exposure. Third, technologies with greater APC exposure subsequently experienced greater growth in U.S. patenting and larger declines in U.S. patenting by inventors located in Axis and occupied countries. We first describe the composition of the vested patent pool and then show how exposure varies across technologies.

Table 1: Summary Statistics for the Vested Patents

	N	Mean	SD	Min	p25	Median	p75	Max
Forward citations	38,294	2.108	3.909	0	0	1	3	143
Firm assignee (=1)	38,294	0.351	0.477	0	0	0	1	1
Filing year	38,294	1931.4	4.9	1914	1927	1931	1936	1942

Notes: Unit of observation is a patent. Forward citations are weighted citations received from U.S. patents granted through [END YEAR]. Firm assignee equals one if the patent was assigned to a firm. Filing year is the year the underlying patent application was filed with the USPTO.

Country of origin. The vested-patent pool is overwhelmingly composed of patents originating in Axis and occupied countries. Table 2 disaggregates the vested-patent pool by the country of the first inventor. The distribution is heavily concentrated among the Axis countries. Patents from the eleven Axis-classified countries account for 70.5% of the pool (26,998 of 38,294), and Germany alone contributes 22,891 patents, 59.8% of all vested patents and 84.8% of the Axis total. The next-largest Axis contributors, Italy (1,504) and Austria (1,203), are an order of magnitude smaller, and Japan (839) smaller still, reflecting the relative size of these countries' pre-war U.S. patenting.

The second-largest source, at 26.9% of the pool (10,317 patents), is occupied territory rather than the Axis countries proper. These are countries brought under vesting authority once Germany, Italy, or Japan occupied them, per the discussion in Section 2. France alone accounts for 6,647 of these, 17.4% of the entire vested-patent pool. Combining Axis and occupied-territory patents captures 97.4% of the pool (37,315 patents across 30 countries), underscoring that the catalog is overwhelmingly a record of Axis and occupied-country, not merely Axis-national, ownership. The residual categories are small and are excluded from the origin-based outcome splits used later in the paper. Allied and neutral countries contribute just 0.7% (265 patents, mostly Switzerland and the UK), and a small number of patents are attributed to the United States (661) or carry no recorded country (53).

Sectoral composition. The geographic concentration of vested patents did not translate into uniform exposure across technologies because foreign patentees specialized in different fields. We map each patent's U.S. Patent Office classification to the two-digit NBER technology categories of Hall et al. (2001) and group these subcategories within six broad sectors. The sectors are Chemical, Computers & Communications, Drugs & Medical, Electrical & Electronic, Mechanical, and Other. Figure 2 reports APC exposure for each two-digit NBER subcategory, defined as the share of patents in the pre-shock patent universe that were vested by the APC. The figure retains subcategories with no vested

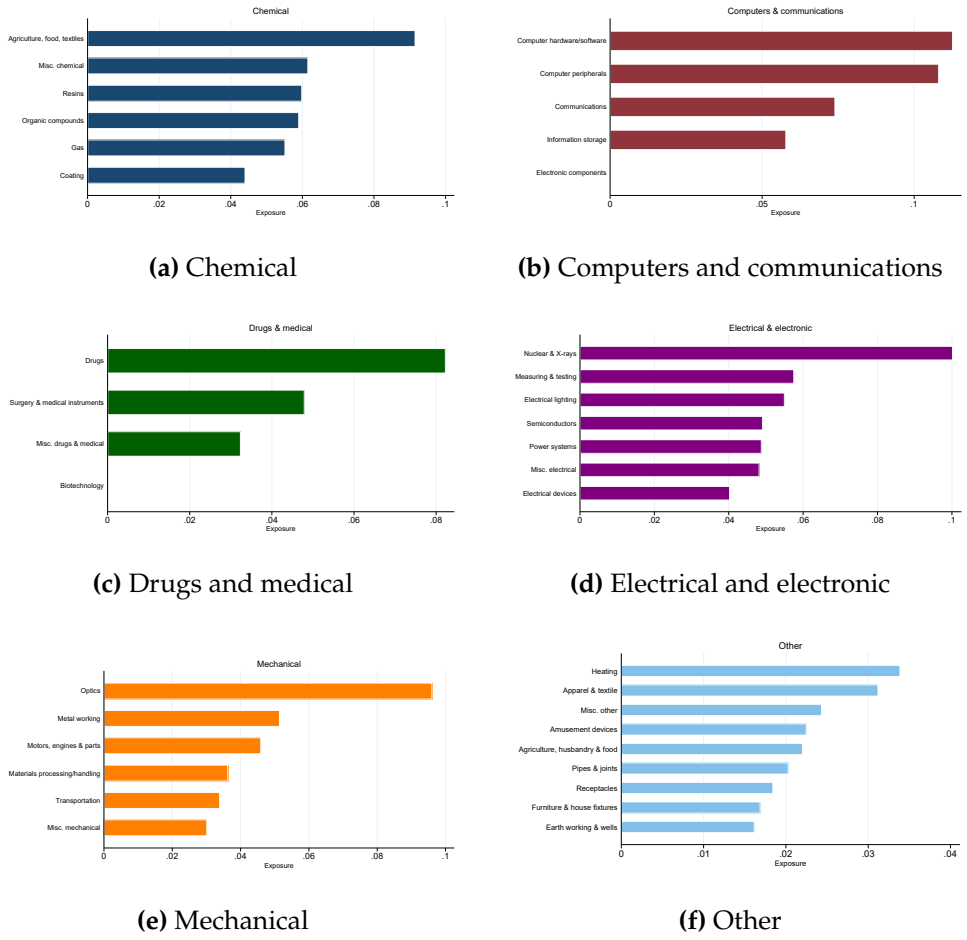
Table 2: Vested patents by country of origin and wartime category

Country	N	Country	N	Country	N
Panel A. Axis (11 countries; N = 26,998)					
Germany	22,891	Hungary	471	Croatia	7
Italy	1,504	Romania	44	Bulgaria	4
Austria	1,203	Slovakia	33	Finland	1
Japan	839			Thailand	1
Panel B. Occupied territory (19 countries; N = 10,317)					
France	6,647	Poland	162	Ukraine	4
Netherlands	957	Indonesia	25	Lithuania	3
Belgium	823	Luxembourg	20	Taiwan	2
Czechoslovakia	616	Greece	10	Slovenia	1
Norway	542	Latvia	10	Korea	1
Denmark	472	Estonia	9		
		Hong Kong	7		
		Serbia	6		
Panel C. Allied and other (25 countries; N = 265)					
Switzerland	107	Australia	8	China	2
United Kingdom	78	Spain	8	Others[†]	17
Canada	17	Russia/USSR	7		
Sweden	14	South Africa	4		
		Mexico	3		
		Ireland	3		
Panel D. Other origins					
United States	661	Missing/blank	53		
Total vested patents				N = 38,294	

Notes: Patents are classified by inventor country of origin (`patent_country`), not assignee nationality. *Axis* denotes enemy nationals; *Occupied* denotes enemy-occupied territory under German, Italian, or Japanese control during WWII. The composite outcome Axis+Occupied sums Panels A and B (N = 37,315, 30 countries). A separate narrow axis definition (*Germany, Japan, Italy, Austria*; N = 26,437) is used for the `patents_axs/patents_ald` outcomes. The United States (N = 661) is reported separately and patents with missing country are excluded from all origin splits. [†]"Others" = Argentina, Bolivia, Chile, Costa Rica, Cuba, Dominican Rep., Egypt, India, Kazakhstan, Nepal, Portugal, Turkey, D.R. Congo, South Sudan (1 each).

patents and assigns them zero exposure. APC exposure varies substantially both across broad sectors and among subcategories within the same sector. Appendix Figures A2–A7 report exposure for each USPTO class within the six sectors, and exposure varies even more across these narrower classes. This uneven distribution provides the cross-technology variation that underlies the empirical analysis.

Figure 2: NBER subcategory-level APC exposure by broad NBER category



(a) Chemical

(b) Computers and communications

(c) Drugs and medical

(d) Electrical and electronic

(e) Mechanical

(f) Other

Notes: Bars report APC exposure for two-digit NBER technology subcategories. Subcategories with no vested patents are retained and assigned exposure equal to zero. Appendix Figures A2–A7 report exposure for each USPTO class within these categories.

Dominant assignees. The uneven distribution of APC exposure across technologies partly reflects the specialized portfolios of major German firms. Table 3 reports 15 selected major assignees, all of them German, together with their principal technology clusters. IG Farbenindustrie AG is the largest, with 1,577 vested patents, or 4.1 percent of the 38,294 matched patents, concentrated in synthetic resins, hydrocarbon chemistry, and dyes.

Telefunken GmbH and Siemens AG account for another 2,362 patents, concentrated in communications and electrical technologies. Together, the three largest assignees account for 3,939 patents, or 10.3 percent of the vested-patent sample.

Table 3: Dominant assignees in vested patents and their technology clusters

Firm	Country	Vested (N)	Dominant technology	NBER two-digit category (USPTO classes)
IG Farbenindustrie AG	Germany	1,577 (4.1%)	Synthetic resins, hydrocarbon chemistry, dyes	Resins [15] (524, 526); Misc. Chemical [19] (423, 585); Agriculture, Food & Textiles [11] (8)
Telefunken GmbH	Germany	1,503 (3.9%)	Radio, antennas, multiplex telephony	Communications [21] (455, 343, 342); Electrical Devices [41] (330, 331, 332)
Siemens AG	Germany	859 (2.2%)	Telephony, electron tubes, electrical systems	Communications [21] (379); Electrical Lighting [42] (313); Measuring & Testing [43] (324)
C. Lorenz AG	Germany	365 (1.0%)	Radar, radio navigation, telecommunications	Communications [21] (342, 343, 455); Electrical Lighting [42] (313)
Carl Zeiss	Germany	260 (0.7%)	Optics, photography, precision instruments	Optics [54] (359); Measuring & Testing [43] (356); Misc. Drugs & Medical [39] (351)
Robert Bosch	Germany	260 (0.7%)	Internal combustion, automotive electrics	Motors, Engines & Parts [53] (123); Power Systems [45] (310)
J. C. Muller & Co.	Germany	166 (0.4%)	Cigarette and tobacco-processing machinery	Agriculture, Husbandry & Food [61] (131); Materials Processing & Handling [51] (53, 493)
Daimler-Benz AG	Germany	139 (0.4%)	Engines, motor vehicles, vehicle components	Motors, Engines & Parts [53] (123); Transportation [55] (180)
MAN AG	Germany	136 (0.4%)	Engines, heavy machinery, hydraulic engineering	Motors, Engines & Parts [53] (123); Earth Working & Wells [64] (405); Gas [13] (48)
Fernseh AG/GmbH	Germany	131 (0.3%)	Television cameras, receivers, studio equipment	Electrical Lighting [42] (313); Misc. Electrical & Electronic [49] (348)
Askania Werke AG	Germany	123 (0.3%)	Measuring, navigation, and aviation instruments	Measuring & Testing [43] (73); Misc. Others [69] (33); Transportation [55] (244)
Rheinmetall AG	Germany	116 (0.3%)	Firearms, ordnance, explosive devices	Misc. Mechanical [59] (89); Misc. Chemical [19] (102)
Vereinigte Stahlwerke AG	Germany	115 (0.3%)	Steel, metallurgy, metalworking	Metal Working [52] (420, 148, 72); Earth Working & Wells [64] (405)
Degussa AG	Germany	112 (0.3%)	Industrial chemistry, precious metals, catalysts	Misc. Chemical [19] (423); Organic Compounds [14] (562, 568); Metal Working [52] (148)
Krupp AG	Germany	106 (0.3%)	Metallurgy, metal treatment, heavy engineering	Metal Working [52] (420, 148)

Top three firms = 3,939 patents (10.3% of all 38,294 matched vested patents).

Notes: The table reports 15 selected major legal assignees. Counts consolidate reviewed spelling, OCR, and historical legal-name variants of the same entity. Separately incorporated subsidiaries, predecessors, joint ventures, and patent-management vehicles are not added to these legal-entity counts; their relationships are retained in the corporate-network data. Percentages are shares of all 38,294 matched vested patents. The final column reports the NBER two-digit technology categories of each firm's leading USPTO classes. NBER category codes appear in square brackets and USPTO class numbers in parentheses.

Firm concentration is considerably sharper within individual USPTO classes. Table 4 lists the classes in which one of these firms is the largest assignee and holds at least a third of the vested patents. It includes only classes with at least 40 vested patents. IG Farbenindustrie accounts for 65.6 percent of vested patents in synthetic resins (class 526) and 64.0 percent in hydrocarbon compounds (class 585). Telefunken holds 64.7 percent in radio-wave antennas (class 343), 62.7 percent in modulators (class 332), 61.3 percent

in oscillators (class 331), and 61.1 percent in active nonlinear electrical circuits (class 327). Thus, even though no single firm dominates the vested pool as a whole, individual technology classes can be closely associated with the portfolios of particular foreign firms.

Table 4: USPTO classes with concentrated vested-patent ownership

Class (NBER two-digit category)	Technology	Vested (N)	Leading firm	Firm share
526 (Resins [15])	Addition polymerization of synthetic resins	61	IG Farbenindustrie AG	65.6%
343 (Communications [21])	Radio-wave antennas	173	Telefunken GmbH	64.7%
585 (Miscellaneous Chemical [19])	Hydrocarbon compounds	161	IG Farbenindustrie AG	64.0%
332 (Electrical Devices [41])	Modulators	142	Telefunken GmbH	62.7%
331 (Electrical Devices [41])	Oscillators	173	Telefunken GmbH	61.3%
327 (Electrical Devices [41])	Active nonlinear electrical circuits	95	Telefunken GmbH	61.1%
455 (Communications [21])	Telecommunications	302	Telefunken GmbH	56.0%
330 (Electrical Devices [41])	Amplifiers	190	Telefunken GmbH	53.7%
518 (Miscellaneous Chemical [19])	Fischer–Tropsch processes and products	46	IG Farbenindustrie AG	50.0%
379 (Communications [21])	Telephonic communications	321	Siemens AG	49.5%
351 (Miscellaneous Drugs and Medical [39])	Eye examination and vision optics	46	Carl Zeiss	47.8%
342 (Communications [21])	Radar and radio navigation	240	Telefunken GmbH	44.6%
536 (Organic Compounds [14])	Organic compounds and carbohydrates	133	IG Farbenindustrie AG	41.4%
333 (Communications [21])	Wave transmission lines and networks	139	Telefunken GmbH	41.0%
525 (Resins [15])	Synthetic resins and natural rubbers	88	IG Farbenindustrie AG	39.8%
363 (Power Systems [45])	Electric power conversion systems	64	Telefunken GmbH	39.1%
131 (Agriculture, Husbandry and Food [61])	Tobacco and cigarette machinery	153	J. C. Muller & Co.	38.6%
358 (Communications [21])	Facsimile and static presentation processing	43	Telefunken GmbH	37.2%
524 (Resins [15])	Synthetic resin and rubber compositions	113	IG Farbenindustrie AG	36.3%
375 (Communications [21])	Pulse or digital communications	47	Telefunken GmbH	36.2%
348 (Miscellaneous Electrical and Electronic [49])	Television	169	Telefunken GmbH	36.1%

Notes: The table reports USPTO classes containing at least 40 matched vested patents in which one of the 15 selected major legal assignees is the leading primary assignee and holds at least one-third of vested patents. Assignee names consolidate reviewed spelling, OCR, and legal-name variants of the same entity; separately incorporated affiliates and corporate predecessors are not consolidated. *Vested* is the total number of matched vested patents in the class. *Firm share* is the leading legal entity's share of those patents. Brackets report NBER two-digit technology categories. Technology descriptions are concise versions of the corresponding USPTO class titles.

Figure 1 illustrates the breadth of the technologies affected by vesting. For each of four NBER subcategories with clear estimated effects, the figure displays the vested patent with the largest recorded forward-citation count. The examples range from amplifier circuits and electron tubes to internal-combustion-engine controls and photographic apparatus. They provide concrete examples of the inventions underlying the sector-level exposure measures while showing that technologically consequential vested patents were not confined to a single industry.

Exposure and subsequent patenting. Appendix Figure A8 provides descriptive evidence on the relationship between subclass-level APC exposure and subsequent changes in patenting by U.S. and foreign-origin patentees. More-exposed subclasses experienced larger increases in U.S. patenting and larger declines in patenting by inventors from Axis and occupied countries. By contrast, the relationship is nearly flat for patents originating in Allied and neutral countries.

5 Empirical Strategy

5.1 Event-Study Design

We estimate event-study models that compare changes in patenting across technologies with and without exposure to APC-vested patents. Let j index USPC class–subclass technology cells and let t index application filing years. [11] The outcome y_{jt} is an annual patent count in cell j, defined separately for each group of patentees considered in the analysis. Our principal outcomes are total patent filings, firm-assigned filings, U.S.-origin filings, and U.S.-origin firm-assigned filings. We also estimate the model for patents originating in Axis and enemy-occupied countries and for patents originating in Allied and other foreign countries.

The baseline treatment, D_j, is the indicator for any APC exposure defined in equation (2). The estimation sample covers 1930–1975 and is restricted to class–subclass cells containing at least seven patents in the pre-APC patent universe. Seven is the median number of patents per subclass in the pre-World War II period (1920–1942); this threshold excludes cells too sparse to identify a meaningful before/after comparison.[12] We estimate the following model by Poisson pseudo maximum likelihood:

$$\mathbb{E}\left[y_{jt} \mid D_j, \alpha_j, \delta_{g(j)t}, N_j^0\right] = \exp\left[\alpha_j + \delta_{g(j)t} + \gamma_t \ln N_j^0 + \sum_{\tau \neq 1938} \beta_\tau D_j \mathbf{1}\{t = \tau\}\right], \qquad (3)$$

where α_j denotes class–subclass fixed effects and $\delta_{g(j)t}$ denotes broad NBER technology-category-by-year fixed effects. The latter allow the six broad technology categories to follow different annual patenting paths. The term $\gamma_t \ln N_j^0$ allows patenting trends to vary flexibly

[11] E.g., class 526, resins; subclass 526/340, interpolymerized with an aromatic hydrocarbon — the class-subclass pair is one technology cell; Section 3.2.

[12] Results are robust to alternative thresholds of 14 and 26 patents (see robustness appendix).

with the log size of the pre-APC patent universe in each cell. The year 1938 is the omitted reference year. Standard errors are clustered at the USPC class level, allowing arbitrary correlation across subclasses and years within a class.

The coefficients β_τ trace the proportional difference in expected patent counts between APC-exposed and unexposed technology cells in year τ, relative to the corresponding difference in 1938. Because 1938 is the omitted reference year, $\beta_{1938} \equiv 0$ by construction: each β_τ measures how much the exposed–unexposed gap in year τ has widened or narrowed relative to that same gap in 1938, not the level of patenting in either group. We report $100[\exp(\beta_\tau) - 1]$ when interpreting an estimate as a percentage change. Coefficients before APC vesting provide a diagnostic for differential pre-treatment trends, while coefficients during and after the war describe the timing and persistence of the divergence in patenting.

We use PPML because the dependent variables are non-negative counts and many class–subclass-year cells contain zero patents. PPML accommodates these zeros without adding an arbitrary constant and remains consistent under general forms of conditional variance misspecification when the conditional mean is correctly specified (Gourieroux et al., 1984; Santos Silva and Tenreyro, 2006). In addition, Chen and Roth (2024) show that treatment effects estimated using log-like transformations, including $\log(1 + y)$ and the inverse hyperbolic sine, can depend on the units in which the outcome is measured when treatment affects the extensive margin. We nevertheless report inverse-hyperbolic-sine and log specifications as robustness checks.

5.2 Alternative Exposure Measures and Specifications

The indicator D_j compares all technology cells containing at least one vested patent with cells containing none. As an alternative, we replace D_j with the continuous APC exposure measure in equation (1), standardized across technology cells within the estimation sample. The resulting coefficients measure the change associated with a one-standard-deviation increase in APC exposure. This specification uses variation in the intensity of vesting, rather than only its presence.

We assess the robustness of the baseline results in several additional ways. We estimate models using citation-weighted patent counts and inverse-hyperbolic-sine transformations of the outcomes. We add the class–subclass OSRD exposure rate, interacted with filing year, to account for the possibility that wartime research funding generated differential patenting growth in the same technologies. We also reconstruct the analysis using CPC main groups as a finer technology classification. These exercises test sensitivity to the measurement of

inventive output, the treatment of zero outcomes, contemporaneous wartime research, and the definition of technology boundaries.

5.3 Identification

The design exploits differences in APC exposure across narrowly defined technology cells. Class–subclass fixed effects absorb permanent differences in patenting levels across technologies, while broad technology-category-by-year fixed effects absorb shocks shared by technologies within the same broad sector in a given year. The year-specific controls for pre-APC cell size further allow larger and smaller technology fields to follow different patenting paths. Identification therefore comes from differential changes in patenting within broad technology categories between more narrowly defined cells with and without vested patents.

The identifying assumption is that, conditional on these fixed effects and controls, exposed and unexposed technology cells would have followed parallel patenting trends in the absence of APC vesting and the associated diffusion of foreign technology. This assumption is not guaranteed by the institutional setting. APC exposure reflected the pre-war technological specialization and U.S. patenting decisions of foreign firms, so exposed fields may have differed from unexposed fields in ways that also predicted subsequent growth. We use the pre-treatment event-study coefficients to assess whether patenting trends were already diverging before the war.

Several additional comparisons help assess competing explanations. Controlling for OSRD exposure addresses differential wartime research investment. Results using continuous exposure test whether the response increases with the intensity of vesting. CPC-based estimates test whether the results depend on the historical USPC boundaries. Finally, separating U.S.-origin patenting from Axis, occupied-country, and other foreign patenting helps distinguish growth in domestic inventive activity from a mechanical change in the composition of foreign filings. These exercises cannot establish that exposure was randomly assigned, but they provide evidence on whether the observed postwar divergence is consistent with the proposed technology-diffusion mechanism.

6 Main Results

We begin by examining how APC exposure changed the national composition of patenting within narrowly defined technologies. We then turn to the paper's main outcomes, total

and firm-assigned patenting by U.S.-origin inventors. The effect on domestic innovation is theoretically ambiguous. Severing links with foreign inventors and firms could have disrupted productive collaborations. Conversely, access to vested intellectual property could have enabled U.S. inventors and firms to build on previously restricted foreign technologies.

6.1 Foreign Patenting in APC-Exposed Technologies

Figure 3 reports estimates of equation (3) for patents originating outside the United States. The first two panels examine patents from Axis and enemy-occupied countries, first for all assignee types and then for patents assigned to firms. The remaining panels report the corresponding outcomes for Allied and other foreign countries.

Patenting from Axis and occupied countries falls sharply in APC-exposed technologies as the United States enters the war. The decline is larger for the outcome covering all assignee types, but it is also pronounced among firm-assigned patents. Some wartime coefficients cannot be estimated because positive patent counts from these countries become extremely sparse in exposed cells. When the coefficients become estimable again after the war, they remain large and negative through 1975. For all Axis and occupied-origin patents, the later estimates are approximately -1.5 log points, which corresponds to a decline of roughly 78 percent relative to the 1938 difference between exposed and unexposed cells. The firm-only estimates are closer to -1 log points, corresponding to a decline of about 63 percent.

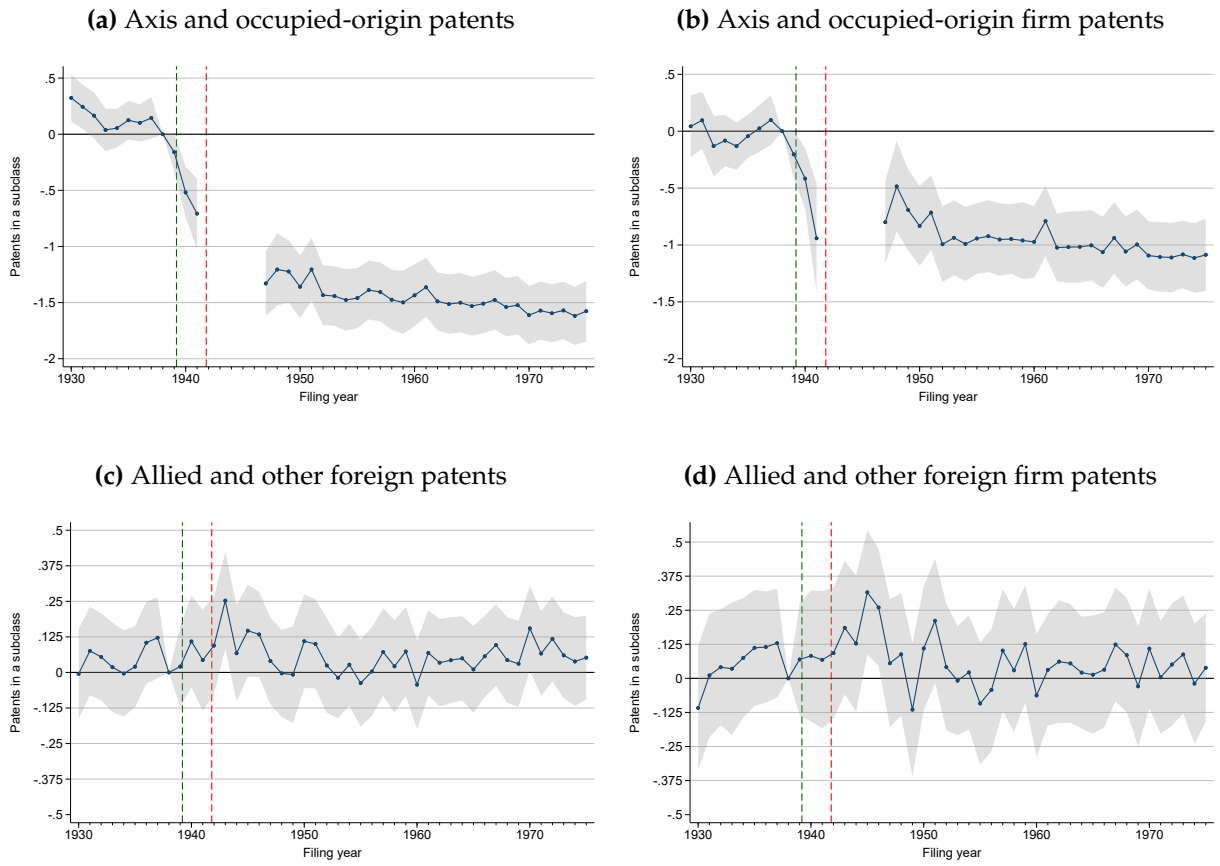
This persistent decline is specific to countries directly affected by the wartime disruption. Patenting from Allied and other foreign countries remains approximately flat, with comparatively imprecise annual estimates. The contrast indicates that the foreign decline is not a general contraction of non-U.S. patenting in APC-exposed technologies.

6.2 U.S. Patenting in APC-Exposed Technologies

Figure 4 presents the main results. The top row reports total and firm-assigned U.S. patent filings regardless of inventor origin. The bottom row restricts the outcomes to U.S.-origin inventors. This distinction matters because total patenting combines the domestic response with the simultaneous decline in patenting from Axis and occupied countries.

The U.S.-origin outcomes show an immediate increase around 1942 in exposed class–subclass cells. The initial rise attenuates somewhat during the late 1940s, but the estimates

Figure 3: APC exposure and foreign-origin patenting



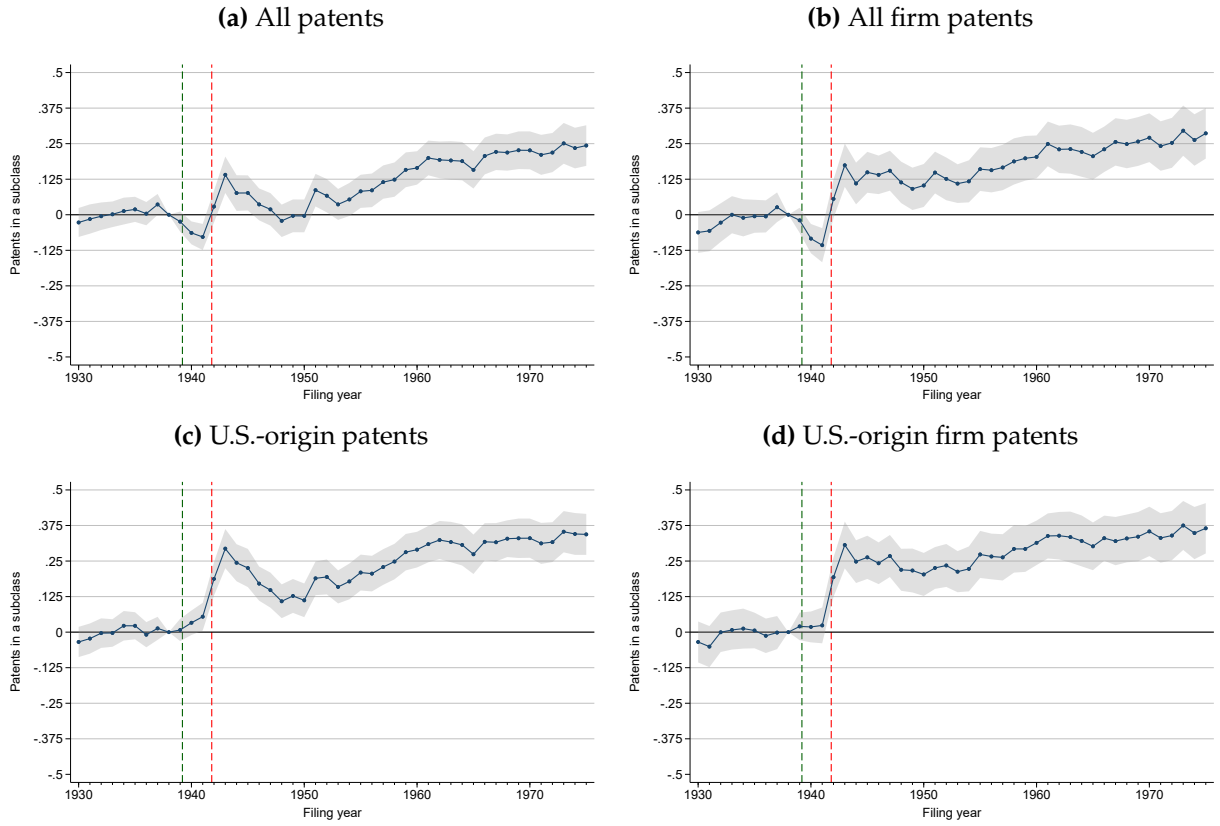
Notes: The figure reports annual PPML estimates of equation (3). The unit of observation is a USPC class–subclass-year cell, treatment indicates any APC exposure, and 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. Missing wartime estimates in panels (a) and (b) reflect coefficients that are not identified because positive Axis and occupied-origin patent counts are extremely sparse.

remain positive and begin increasing again during the 1950s. By the 1960s and early 1970s, the coefficients for U.S.-origin and U.S.-origin firm patenting are generally between 0.3 and 0.4, implying approximately 35 to 49 percent higher expected patent counts in exposed cells, relative to the corresponding difference in 1938. The firm outcome displays an especially clear and persistent response, consistent with U.S. firms being important beneficiaries of access to vested foreign technologies.

The results for patenting regardless of inventor origin are more muted around the war. Total patenting declines before 1942 and displays a smaller initial increase because the domestic response is partly offset by the collapse in Axis and occupied-origin filings documented in Figure 3. Nevertheless, both total and firm-assigned patenting become persistently positive in the postwar period and rise further after the early 1950s. Taken

together, the two figures show a lasting change in the composition of patenting within exposed technologies: foreign patenting from the affected countries contracts, while patenting by U.S. inventors and firms expands.

Figure 4: APC exposure and U.S. patenting



Notes: The figure reports annual PPML estimates of equation (3). The unit of observation is a USPC class–subclass-year cell, treatment indicates any APC exposure, and 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. Panels (a) and (b) include patents of all inventor origins; panels (c) and (d) restrict the outcomes to patents whose first-listed inventor is located in the United States. Firm outcomes count patents with at least one firm assignee.

6.3 Robustness

The tables below assess the sensitivity of the main results to alternative outcome and treatment definitions described in Section 5.2. Table 5 compares the baseline any-exposure treatment with standardized continuous APC exposure and compares PPML estimates using count outcomes with OLS estimates using inverse-hyperbolic-sine outcomes. Across these specifications, U.S.-origin patenting rises after 1942 and remains elevated through 1975. The estimates therefore do not depend on using a binary treatment measure or PPML alone. Because the tables group years into periods and use 1930–1938 as the reference

period, their coefficients should not be compared mechanically with individual annual coefficients in the event-study figures.

Table 5: APC exposure and U.S. patenting: OLS and PPML estimates

	OLS: IHS outcomes				PPML: count outcomes			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable	U.S. patents	U.S. firm patents	U.S. patents	U.S. firm patents	U.S. patents	U.S. firm patents	U.S. patents	U.S. firm patents
Treatment	Any	Any	Continuous	Continuous	Any	Any	Continuous	Continuous
APC exposure × 1939–1941	0.013*	0.004	0.004	0.000	0.034*	0.031	0.024*	0.016
	(0.007)	(0.005)	(0.003)	(0.002)	(0.019)	(0.024)	(0.012)	(0.015)
APC exposure × 1942–1945	0.078***	0.045***	0.037***	0.023***	0.239***	0.261***	0.171***	0.173***
	(0.008)	(0.006)	(0.004)	(0.003)	(0.028)	(0.032)	(0.018)	(0.021)
APC exposure × 1946–1948	0.050***	0.041***	0.027***	0.023***	0.147***	0.253***	0.123***	0.171***
	(0.008)	(0.006)	(0.003)	(0.003)	(0.026)	(0.032)	(0.018)	(0.020)
APC exposure × 1949–1959	0.062***	0.042***	0.029***	0.019***	0.197***	0.258***	0.142***	0.160***
	(0.007)	(0.006)	(0.003)	(0.003)	(0.025)	(0.030)	(0.019)	(0.023)
APC exposure × 1960–1975	0.096***	0.062***	0.040***	0.026***	0.322***	0.345***	0.186***	0.183***
	(0.008)	(0.007)	(0.004)	(0.003)	(0.030)	(0.036)	(0.019)	(0.022)
Observations	1,900,444	1,900,444	1,900,444	1,900,444	1,894,418	1,792,850	1,894,418	1,792,850
Dependent-variable mean	0.387	0.244	0.387	0.244	0.592	0.375	0.592	0.375
Method	**OLS**	**OLS**	**OLS**	**OLS**	**PPML**	**PPML**	**PPML**	**PPML**
Subclass FE	×	×	×	×	×	×	×	×
Filing-year FE	×	×	×	×	×	×	×	×
NBER one-digit × year FE	×	×	×	×	×	×	×	×
ln(universe)× year	×	×	×	×	×	×	×	×
SE clustered by USPTO class	×	×	×	×	×	×	×	×

Standard errors in parentheses

Reference period is 1930–1938; the sample requires n_universe >= 7.

Any exposure equals one when APC exposure is positive; continuous exposure is standardized in the estimation sample.

OLS outcomes are inverse-hyperbolic-sine transformed; PPML outcomes are patent counts.

Every column includes subclass and filing-year fixed effects, NBER one-digit category-by-year fixed effects, and ln(universe)-by-year controls.

Standard errors are clustered by USPTO class.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Tables 6 and 7 replace raw patent counts with citation-weighted counts.[13] The estimates remain positive for U.S.-origin patenting and negative for patenting from Axis and occupied countries. In the most saturated specifications, which include broad or two-digit NBER category-by-year effects, filing-year interactions with the log pre-APC patent universe, and OSRD-rate-by-year controls, the postwar coefficients retain the same signs and are statistically distinguishable from zero.

Table 8 adds the OSRD-funded share of each subclass's patenting (Gross and Sampat) alongside APC exposure, both interacted with the same five periods, together with their interaction; the APC exposure coefficients are essentially unchanged, and the APC-by-OSRD interaction is small and statistically indistinguishable from zero throughout, so the postwar rise in U.S.-origin patenting is not driven by wartime research funding. Appendix Table A1 instead interacts APC exposure with the mean remaining term of a subclass's vested patents at the time of vesting; this interaction is positive and, in most periods, statistically distinguishable from zero, and the pattern survives controlling for field age. Subclasses whose vested patents had more years of protection remaining therefore show a larger patenting response, consistent with the removal of patent protection, rather than

[13] We count all citations to the vested patent in the following 20 years.

vesting alone, driving the increase.

7 Sources of the Aggregate Effect

The estimates in Section 6.2 pool all technologies, while the vested portfolios were concentrated in a few (Tables 3 and 4). This section asks where the U.S. response occurred. We first estimate the baseline specification within each of the six NBER sectors. We then estimate it within the two-digit categories in which the leading German assignees held their patents.

Figures 5 and 6 report equation (3) estimated separately within each sector, for U.S.-origin patents and for U.S. firm patents. Three patterns are common to the sectors.

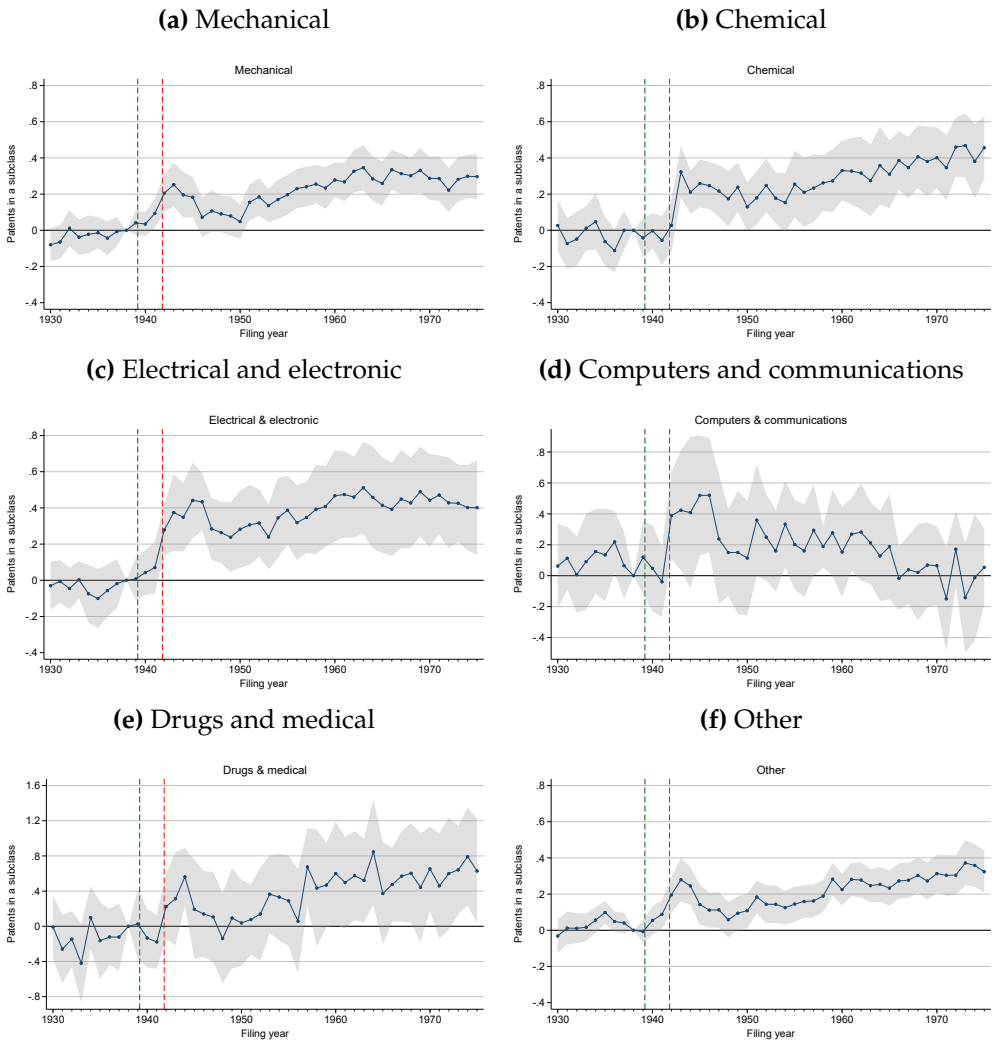
First, timing. Coefficients through 1941 are small in every sector, and the increase occurs between 1942 and 1943, when the Custodian vested and catalogued the patents, rather than at the outbreak of war in Europe or at U.S. entry.

Second, the shape of the wartime response. U.S.-origin patenting rises in every sector during 1942–46 and then attenuates in 1947–50 in all but chemicals. The firm series show little attenuation: in mechanical technologies U.S. firm patenting holds its wartime level through the late 1940s while the U.S.-origin series falls sharply. The reconversion trough belongs to individual inventors; firms kept the wartime gain. The fine-category estimates in Figure 7, 8, and 9 therefore use U.S. firm patents.

Third, the long-run level. By 1960–75 the firm coefficients are 0.4–0.5 in chemicals and still rising, 0.3–0.4 in mechanical and electrical technologies, and 0.25–0.4 in the other sector, implying 35 to 65 percent more patents in exposed cells. Drugs and medical has the largest estimates, 0.5–1.0 after 1957. Computers and communications is the exception: after a wartime peak the coefficient declines through the 1950s and is indistinguishable from zero after the mid-1960s.

The response was therefore general and corporate. Tables 3 and 4 locate the largest vested portfolios: IG Farben in resins and hydrocarbon chemistry; Telefunken, Siemens, and Lorenz in radio, telephony, and circuits; Bosch, Daimler-Benz, and MAN in engines; Krupp and Vereinigte Stahlwerke in metal working; Zeiss in optics; Rheinmetall in ordnance. For each sector we estimate equation (3) within the two-digit NBER categories that contain these classes, with U.S. firm patents as the outcome and the standardized continuous exposure measure of Section 5.2 as treatment. Coefficients are per standard deviation of exposure and are not comparable in level to those in the sector figures.

Figure 5: APC exposure and U.S.-origin patenting by NBER technology category



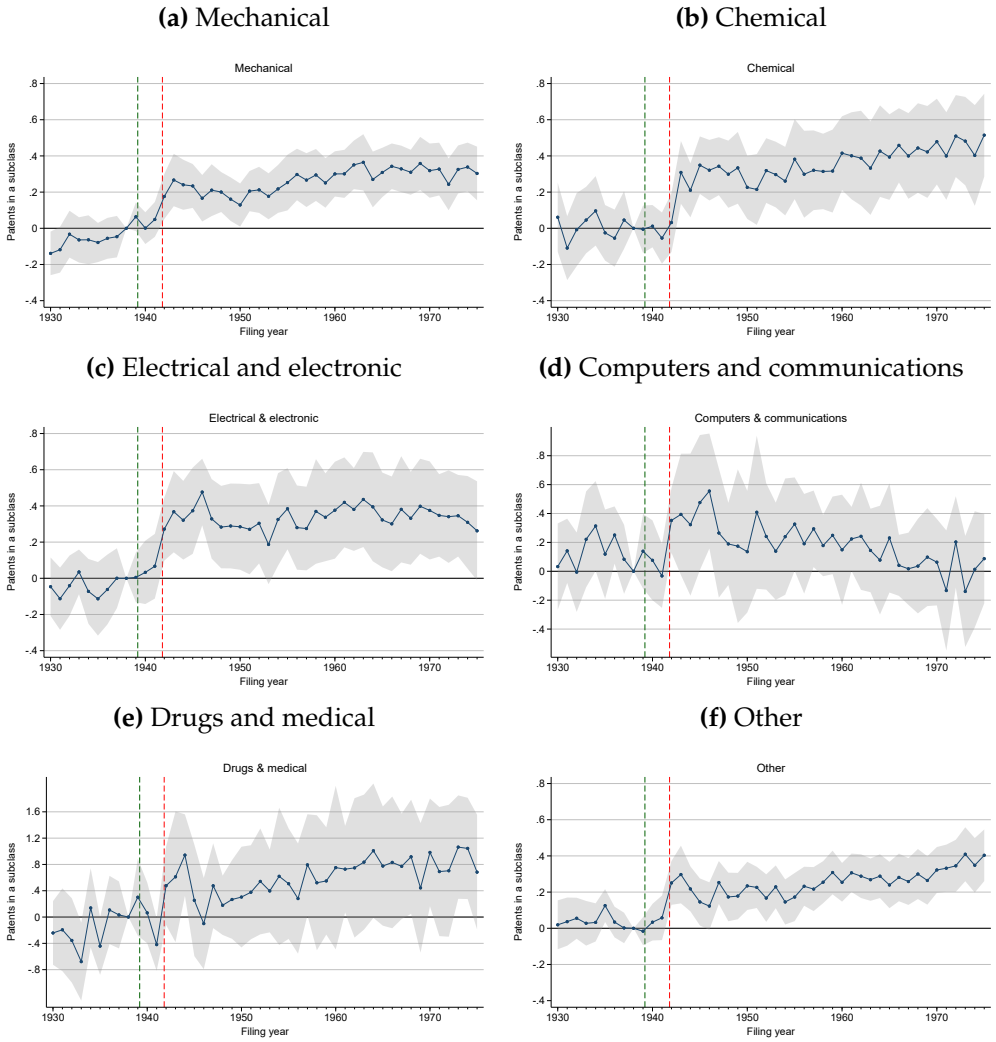
Notes: Each panel reports annual PPML estimates of equation (3) within the indicated broad NBER technology category. The outcome is the number of U.S.-origin patent filings in a USPC class–subclass-year cell. Treatment indicates any APC exposure, and 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. The vertical scale differs in panel (e).

7.1 Mechanical and Engineering Technologies

Mechanical technologies account for a third of vested patents, and panel (a) of the sector figures showed the most stable firm response of any sector. The engineering firms in Table 3 held their patents in six two-digit categories: engines, metal working, earth working, heating, ordnance, and optics.[14] Figure 7 reports the estimates.

[14]Engines (NBER 53; class 123): Daimler-Benz, MAN, Bosch, Maybach; 2,755 vested patents. Metal working (52; classes 420, 148, 72): Vereinigte Stahlwerke, Krupp; 1,801. Earth working (64; class 405): MAN, Vereinigte Stahlwerke; 306. Heating (66; furnaces, burners, heat exchange): Siemens, Krupp, Junkers, Askania; 1,062. Ordnance (59; class 89): Rheinmetall; 2,848. Optics (54; class 359): Zeiss, Leitz; 721. Hall et al.

Figure 6: APC exposure and U.S.-origin firm patenting by NBER technology category



Notes: Each panel reports annual PPML estimates of equation (3) within the indicated broad NBER technology category. The outcome is the number of U.S.-origin patent filings with at least one firm assignee in a USPC class–subclass-year cell. Treatment indicates any APC exposure, and 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. The vertical scale differs in panel (e).

Panel (a), engines, shows the clearest response: near zero through 1941, a rise to about 0.4 (about 45 percent more U.S. firm patents per standard deviation of exposure) in 1943, and no decay through 1975. The Custodian singled out Bosch's ignition systems and Daimler-Benz's automobile inventions among its holdings, and the Bosch condenser machine of Section 2 shows that a patent alone did not always suffice to reproduce a process. The engine estimates show that the patents nonetheless supported a lasting expansion of

(2001) file earth working and heating under Other and Rheinmetall's ammunition class 102 under chemicals; we group them with the engineering firms that held them.

U.S. firm invention.

Panel (b), metal working, shows the smallest response, a coefficient of 0.10–0.15 from the war onward, or roughly 11 to 16 percent higher expected patent counts relative to 1938. Panels (c) and (d), earth working and heating, share a different shape: both decline through the late 1930s, rise after 1942, and settle at high levels, about 0.6 for earth working and 0.8 for heating, the largest in the figure, corresponding to increases of roughly 82 and 123 percent. Panel (e), ordnance, is the one mechanical category with a wartime peak; the coefficient reaches 0.45 in 1945 (about 57 percent), falls to 0.1 by 1950 (about 11 percent), and settles near 0.3 (about 35 percent). Panel (f), optics, rises during the war and holds 0.2–0.35 through 1975, or roughly 22 to 42 percent. Within mechanical and engineering technologies, then, the response is immediate and stable in engines, ordnance, and optics; gradual and largest in heating and earth working; and smallest in metal working.

7.2 Electrical, Electronic, and Communications Technologies

Electrical and electronic technologies account for 14 percent of vested patents and communications for 6 percent. Panels (c) and (d) of the sector figures showed the largest wartime responses. The electrical response stays high through 1975; the communications response declines and is near zero after the mid-1960s.

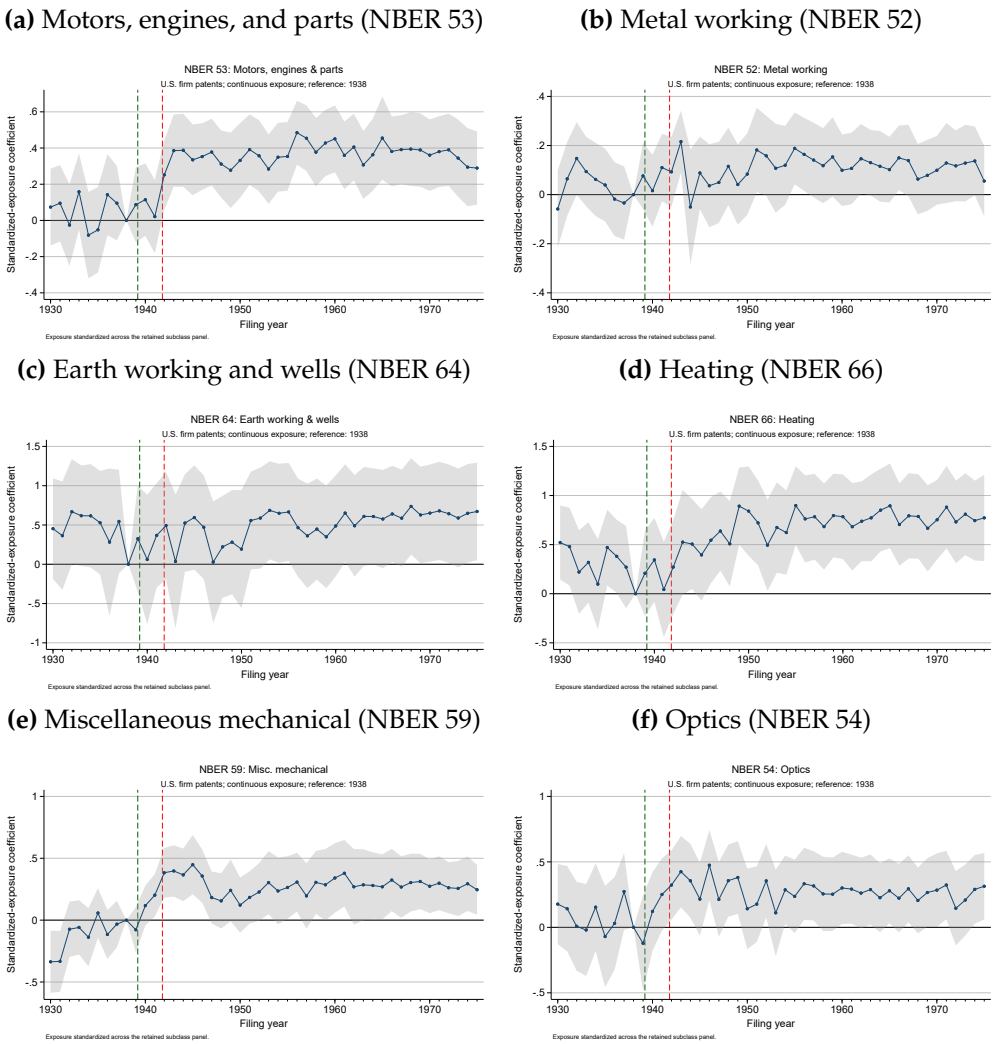
Two firms dominate both groups. Telefunken and Siemens are the two largest holders in five of the six categories examined here: electron tubes, power systems, communications, circuits, and television and electric heating.[15] Lorenz is third in circuits, communications, and tubes; Fernseh in tubes and television. The sixth, measuring and testing, is held by Siemens, Zeiss, and Askania.[16] Communications and circuits are the most concentrated categories in the vested set: Table 4 shows Telefunken holding 36 to 65 percent of the vested patents in six communications classes and 53 to 63 percent in the four circuit classes, and Siemens half of telephony. Figure 8 reports the estimates.

Panel (a), electron tubes, shows the largest and most immediate response: the coefficient rises to 0.5 (about 65 percent more U.S. firm patents per standard deviation of exposure) by 1943 and stays between 0.4 and 0.6 (roughly 49 to 82 percent) through 1975. Panel (b), power systems, has the same shape at a lower level, 0.3–0.4 (roughly 35 to 49 percent) from 1946. Panel (c), which pools the measuring and testing classes 73 and 324, rises to 0.58 (about 79

[15] Electron tubes (NBER 42; class 313): 1,044 vested patents. Power systems (45; classes 310 and 60): 1,144. Communications (21; classes 379, 455, 342, 343, 333): 1,574. Circuits (41; classes 330, 331, 332, 327): 1,430. Television and electric heating (49; classes 348 and 219): 725.

[16] Measuring and testing: NBER 43; classes 73, 324, 356; 816 vested patents.

Figure 7: APC exposure and U.S. firm patenting within mechanical and engineering technologies



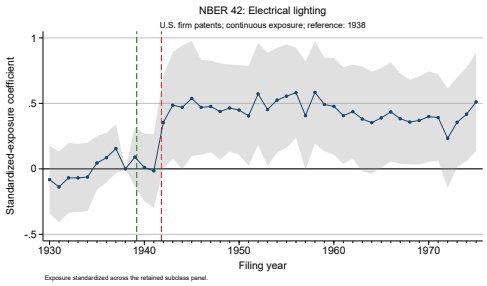
Notes: Each panel reports annual PPML estimates of equation (3) within the indicated two-digit NBER technology category. The outcome is the number of U.S.-origin patent filings with at least one firm assignee in a USPC class–subclass-year cell. Treatment is the continuous APC exposure measure of equation (1), standardized within the estimation sample, so coefficients are per standard deviation of exposure; 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. The vertical scale differs across panels.

percent) in 1943 and holds 0.3–0.5 (roughly 35 to 65 percent) through the mid-1960s before declining. Panel (d), communications, rises to 0.35 (about 42 percent) by 1945, holds about 0.2 (roughly 22 percent) through the early 1960s, and declines thereafter. The two small categories that complete the communications sector are not shown. Computer hardware (NBER 22), which in this period is almost entirely class 396, photography, shows no response; information storage (24) returns to zero by 1975. The fade in Figure 6(d) combines these three paths. Panels (e) and (f), circuits and miscellaneous electrical, show the smallest

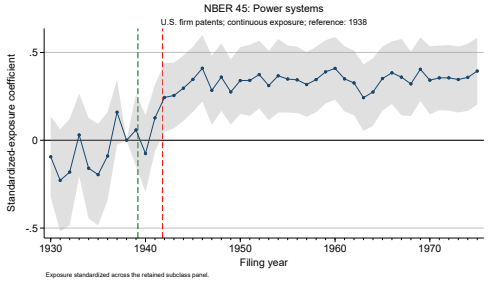
responses, 0.1–0.2 (roughly 11 to 22 percent) throughout. Within these technologies, then, the response is largest in electron tubes, measuring and testing, and power systems; positive but declining in communications; and smallest in the circuit classes where Telefunken's ownership was most concentrated.

Figure 8: APC exposure and U.S. firm patenting within electrical, electronic, and communications technologies

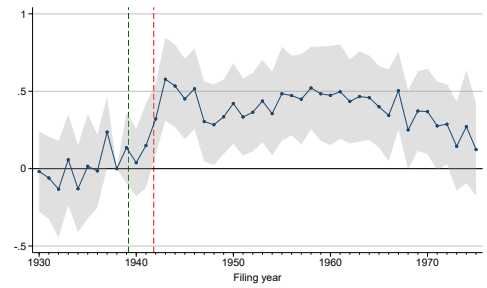
(a) Electron tubes (NBER 42, electrical lighting)



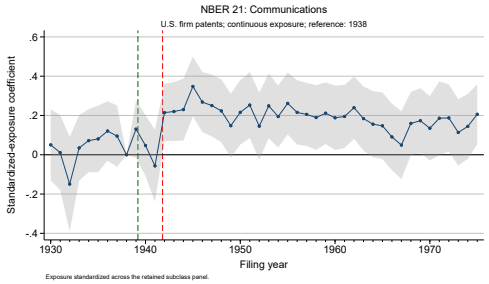
(b) Power systems (NBER 45)



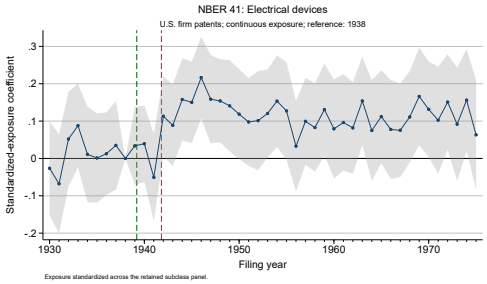
(c) Measuring and testing (classes 73 and 324)



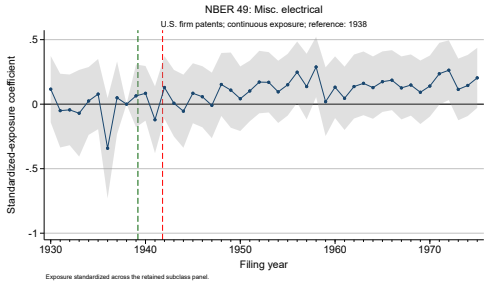
(d) Communications (NBER 21)



(e) Circuits (NBER 41, electrical devices)



(f) Miscellaneous electrical (NBER 49)



Notes: Each panel reports annual PPML estimates of equation (3) within the indicated two-digit NBER technology category; panel (c) pools USPC classes 73 and 324, mechanical and electrical measuring and testing. The outcome is the number of U.S.-origin patent filings with at least one firm assignee in a USPC class–subclass-year cell. Treatment is the continuous APC exposure measure of equation (1), standardized within the estimation sample, so coefficients are per standard deviation of exposure; 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. The vertical scale differs across panels.

7.3 Chemical Technologies

Chemical technologies account for 21 percent of vested patents, and panel (b) of the sector figures showed the only response still rising in the 1970s. IG Farben's 1,577 vested patents (Table 3) fall largely in five two-digit categories: miscellaneous chemical, organic compounds, resins, agriculture, food, and textiles, and coating.[17] Degussa, Kalle, Wacker, and Ruhrchemie follow at a distance, and MAN is the second holder in gas.[18] In Table 4, IG Farben holds 36 to 66 percent of the vested patents in the resin and hydrocarbon classes, the highest concentration of any firm in the vested set. Figure 9 reports the estimates.

Panel (a), miscellaneous chemical, carries the sector result: the coefficient rises to 0.44 (about 55 percent) by 1946 and to 0.4–0.5 (roughly 49 to 65 percent) in 1960–75, still increasing at the end of the sample. Panel (b), coating, is the highest in the figure, holding 0.45–0.6 (roughly 57 to 82 percent) from 1943. Panels (c) and (d), agriculture, food, and textiles and gas, hold about 0.3 (roughly 35 percent) through the 1950s; gas then declines to 0.1 (about 11 percent) by 1975. Panels (e) and (f) are the exception. Organic compounds and resins are the categories in which IG Farben's ownership was most concentrated, yet their responses are the smallest in the figure: both jump in 1942 and stay flat, at 0.1–0.2 (roughly 11 to 22 percent) and about 0.1 (about 11 percent), through 1975. Access in these fields probably pre-dated vesting. IG Farben's synthetic rubber and polymer patents had been licensed to Standard Oil of New Jersey under the firms' 1929–30 agreements, and the 1942 antitrust decree and the Rubber Reserve program made them available to American producers on terms the Custodian's licenses did little to improve (Borkin, 1978). Within chemical technologies, then, the response is largest and still rising where IG Farben's holdings were largest, and smallest where its ownership share was highest.

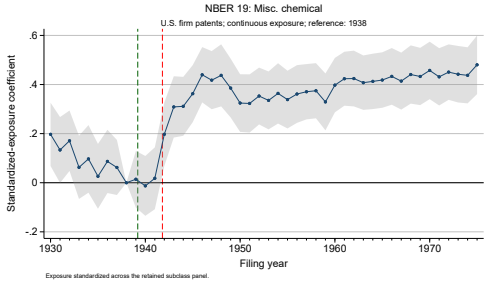
Taken together, the fine-category estimates place the aggregate response where the vested portfolios were: engines, electron tubes, power systems, and industrial chemistry, the classes of Bosch, Daimler-Benz, Telefunken, Siemens, and IG Farben. The exceptions are resins and organic compounds, where access pre-dated vesting, and communications, where the response declines after the early 1960s.

[17] Miscellaneous chemical (NBER 19; classes 423, 585, 208): 675 IG Farben patents of 4,803 vested. Organic compounds (14): 271 of 1,014. Resins (15): 168 of 460. Agriculture, food, and textiles (11; mainly class 8, dyeing): 100 of 614. Coating (12): 66 of 727.

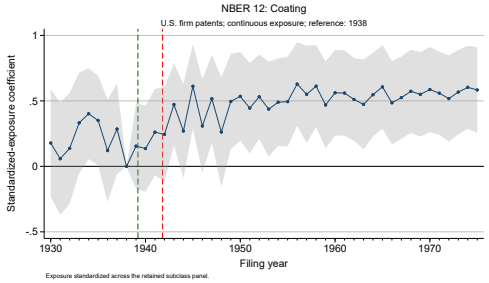
[18] Gas: NBER 13; 311 vested patents.

Figure 9: APC exposure and U.S. firm patenting within chemical technologies

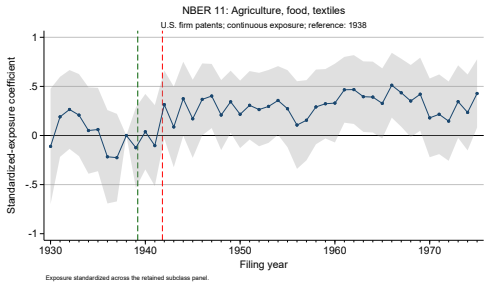
(a) Miscellaneous chemical (NBER 19)



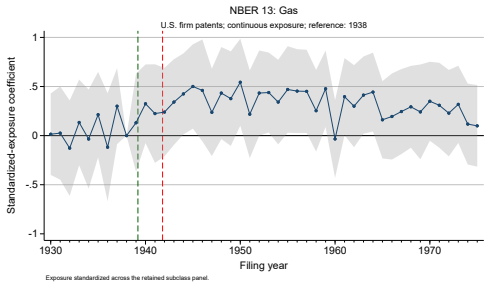
(b) Coating (NBER 12)



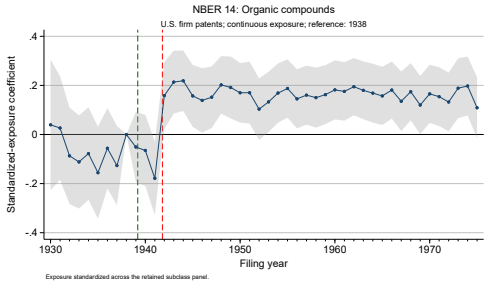
(c) Agriculture, food, and textiles (NBER 11)



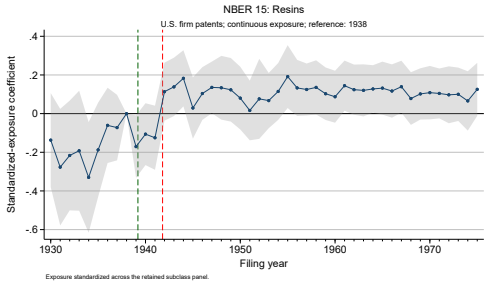
(d) Gas (NBER 13)



(e) Organic compounds (NBER 14)



(f) Resins (NBER 15)



Notes: Each panel reports annual PPML estimates of equation (3) within the indicated two-digit NBER technology category. The outcome is the number of U.S.-origin patent filings with at least one firm assignee in a USPC class–subclass-year cell. Treatment is the continuous APC exposure measure of equation (1), standardized within the estimation sample, so coefficients are per standard deviation of exposure; 1938 is the omitted year. Shaded areas report 95 percent confidence intervals based on standard errors clustered by USPC class. The vertical scale differs across panels.

8 Conclusion

This paper studies a forced technology transfer, the seizure of foreign-owned intellectual property in the United States during World War II. The Alien Property Custodian took title to the U.S. patents of residents of the Axis countries and of the countries they occupied, and it licensed them to any American applicant. We digitize the Custodian's catalog of the vested patents, link it to the historical patent record, and compare patenting across

technology subclasses that the seizure reached to different degrees. U.S.-origin patenting rose in exposed subclasses in 1942, the year the patents were vested, and remained higher through 1975, by 27 percent in 1942 to 1945 and by 38 percent in 1960 to 1975. The estimates hold under alternative measures of exposure and of patenting, and they are not explained by wartime research funding. Patenting by inventors located in Axis and occupied countries fell in the same subclasses by about 78 percent and did not recover.

The timing differs from that of the earlier episode for which there is evidence, the licensing of German chemical patents after World War I. There the largest response of U.S. patenting came six to nine years after licensing. Moser and Voena (2012) attribute the lag to the time American firms needed to learn to produce the German inventions, in an industry that was then new to the United States. After World War II the response was immediate. U.S.-origin patenting rose in exposed subclasses in 1942, the year the patents were vested. Two differences in the setting may account for this. First, American industry was more mature than it had been after World War I. Second, the seizure took place in the middle of a war. American manufacturers were producing for it, and the Custodian catalogued and licensed the patents for their immediate use in war production.

The response was large and lasting where the vested holdings were large, in engines, electron tubes, power systems, and industrial chemistry, and it was larger in subclasses whose vested patents had more years of protection remaining. It was small in synthetic resins and organic compounds, the fields in which IG Farben's ownership was most concentrated but in which American producers had probably obtained access to the technology before the war. Both patterns are consistent with an effect that worked through the patents themselves. The seizure was not, however, the only measure taken against the owners. Congress barred the return of their property, the United States collected know-how from German plants after 1945, and the Allies broke up IG Farben. Our estimates capture the seizure together with these measures. The decline in patenting by inventors located in Axis and occupied countries began in 1939, before the seizure, and it reflects the war and its aftermath as well as the loss of the patents.

The seizure was an industrial policy of an unusual kind. It did not subsidize production or protect a market. It took an existing stock of foreign technology, in many fields at once, and licensed it to any American applicant without exclusivity and, for Axis-owned patents, without royalty. The results show that such a transfer can raise domestic patenting for decades. Whether they carry over to other settings is less clear. The transfer took place in wartime, and the owners were residents of states with which the United States was at war and had no recourse. A seizure in peacetime would also change the incentives of foreign

inventors to patent in the seizing country, an effect that Moser and Voena (2012) raise for compulsory licensing and that we cannot separate here from the effect of the war itself.

Our estimates concern patenting. We do not measure the effects of the seizure on the output or productivity of the firms that used the patents, or the losses of the owners. The Custodian reported that 1,097 firms and government agencies had taken licenses by June 1946, and identifying these firms in the patent record would show which firms used the patents and what they did with them. The Bosch case in Section 2 shows that a patent did not always suffice to reproduce a process. We leave to future work the question of how much of the response required know-how in addition to the patents.

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Tables

Table 6: Subclass PPML: Citation-weighted U.S. patenting (Any APC exposure)

	CW U.S. patents					CW U.S. firm patents				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
APC exposed x 1939-1941	-0.033	-0.054**	-0.008	-0.002	-0.015	-0.048	-0.065**	-0.020	-0.013	-0.032
	(0.024)	(0.023)	(0.025)	(0.023)	(0.023)	(0.030)	(0.029)	(0.030)	(0.028)	(0.027)
APC exposed x 1942-1945	0.193***	0.151***	0.228***	0.227***	0.197***	0.190***	0.156***	0.226***	0.228***	0.194***
	(0.032)	(0.031)	(0.030)	(0.030)	(0.029)	(0.039)	(0.036)	(0.035)	(0.034)	(0.033)
APC exposed x 1946-1948	0.054*	0.051*	0.121***	0.138***	0.119***	0.120***	0.114***	0.201***	0.208***	0.184***
	(0.032)	(0.030)	(0.029)	(0.028)	(0.027)	(0.039)	(0.037)	(0.037)	(0.035)	(0.034)
APC exposed x 1949-1959	0.084***	0.080***	0.167***	0.187***	0.172***	0.104***	0.108***	0.196***	0.218***	0.202***
	(0.029)	(0.028)	(0.029)	(0.027)	(0.027)	(0.036)	(0.035)	(0.036)	(0.033)	(0.034)
APC exposed x 1960-1975	0.082**	0.083**	0.234***	0.242***	0.233***	0.073*	0.090**	0.236***	0.248***	0.237***
	(0.034)	(0.033)	(0.033)	(0.031)	(0.032)	(0.040)	(0.037)	(0.039)	(0.036)	(0.037)
Observations	1861206	1861206	1861206	1861206	1603008	1741054	1741054	1741054	1741054	1523888

Standard errors in parentheses

Reference period is 1930-1938.

Treatment equals one for subclasses with any APC exposure and zero otherwise.

Outcomes are citation-weighted patent counts, estimated by PPML at the USPTO class-subclass-year level; standard errors are clustered by USPTO class.

All columns include subclass and filing-year fixed effects.

Columns (2)-(3) and (7)-(8) add NBER main-category by year fixed effects; columns (4)-(5) and (9)-(10) use NBER subcategory by year fixed effects.

Columns (3)-(5) and (8)-(10) add ln(universe) by filing-year interactions; columns (5) and (10) also add OSRD-rate by filing-year interactions.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Subclass PPML: Citation-weighted Axis and occupied-origin patenting (Any APC exposure)

	CW Axis + occupied patents					CW Axis + occupied firm patents				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
APC exposed x 1939-1941	-0.420***	-0.439***	-0.479***	-0.466***	-0.420***	-0.257**	-0.271***	-0.335***	-0.318***	-0.286***
	(0.095)	(0.095)	(0.099)	(0.100)	(0.099)	(0.102)	(0.099)	(0.104)	(0.103)	(0.105)
APC exposed x 1942-1945	-1.356***	-1.374***	-1.659***	-1.665***	-1.656***	-0.910***	-0.917***	-1.252***	-1.246***	-1.245***
	(0.170)	(0.173)	(0.172)	(0.169)	(0.169)	(0.268)	(0.271)	(0.269)	(0.266)	(0.264)
APC exposed x 1946-1948	-1.247***	-1.264***	-1.414***	-1.434***	-1.423***	-0.740***	-0.761***	-0.876***	-0.903***	-0.904***
	(0.130)	(0.130)	(0.130)	(0.122)	(0.120)	(0.155)	(0.156)	(0.149)	(0.138)	(0.137)
APC exposed x 1949-1959	-1.366***	-1.354***	-1.488***	-1.529***	-1.500***	-0.838***	-0.824***	-0.951***	-1.012***	-0.999***
	(0.093)	(0.093)	(0.096)	(0.092)	(0.093)	(0.102)	(0.101)	(0.106)	(0.101)	(0.103)
APC exposed x 1960-1975	-1.565***	-1.538***	-1.673***	-1.711***	-1.682***	-1.066***	-1.035***	-1.160***	-1.216***	-1.192***
	(0.092)	(0.092)	(0.096)	(0.090)	(0.089)	(0.099)	(0.098)	(0.102)	(0.094)	(0.094)
Observations	1145630	1143919	1143919	1121298	1012037	848838	843355	843355	809935	738659

Standard errors in parentheses

Reference period is 1930-1938.

Treatment equals one for subclasses with any APC exposure and zero otherwise.

Outcomes are citation-weighted patent counts, estimated by PPML at the USPTO class-subclass-year level; standard errors are clustered by USPTO class.

All columns include subclass and filing-year fixed effects.

Columns (2)-(3) and (7)-(8) add NBER main-category by year fixed effects; columns (4)-(5) and (9)-(10) use NBER subcategory by year fixed effects.

Columns (3)-(5) and (8)-(10) add ln(universe) by filing-year interactions; columns (5) and (10) also add OSRD-rate by filing-year interactions.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Subclass PPML: U.S. patenting, continuous APC exposure and OSRD rate

	U.S. patents		U.S. firm patents	
	(1)	(2)	(3)	(4)
APC exposure				
APC exposure x 1939-1941	0.019	0.022*	0.011	0.014
	(0.012)	(0.012)	(0.015)	(0.015)
APC exposure x 1942-1945	0.154***	0.159***	0.156***	0.161***
	(0.016)	(0.016)	(0.019)	(0.019)
APC exposure x 1946-1948	0.113***	0.123***	0.163***	0.169***
	(0.016)	(0.015)	(0.019)	(0.018)
APC exposure x 1949-1959	0.129***	0.142***	0.149***	0.159***
	(0.017)	(0.016)	(0.020)	(0.019)
APC exposure x 1960-1975	0.176***	0.183***	0.175***	0.180***
	(0.018)	(0.017)	(0.021)	(0.020)
OSRD rate				
OSRD rate x 1939-1941	0.043**	0.046**	0.041*	0.045**
	(0.020)	(0.020)	(0.022)	(0.021)
OSRD rate x 1942-1945	0.149***	0.148***	0.131***	0.131***
	(0.026)	(0.025)	(0.027)	(0.027)
OSRD rate x 1946-1948	0.088***	0.091***	0.080***	0.083***
	(0.024)	(0.023)	(0.025)	(0.025)
OSRD rate x 1949-1959	0.113***	0.109***	0.102***	0.098***
	(0.024)	(0.024)	(0.026)	(0.025)
OSRD rate x 1960-1975	0.124***	0.117***	0.111***	0.105***
	(0.025)	(0.024)	(0.025)	(0.025)
APC exposure × OSRD rate				
APC exposure x OSRD rate x 1939-1941	0.007	0.007	0.007	0.008
	(0.015)	(0.014)	(0.017)	(0.016)
APC exposure x OSRD rate x 1942-1945	0.003	0.005	0.002	0.004
	(0.021)	(0.021)	(0.022)	(0.022)
APC exposure x OSRD rate x 1946-1948	0.007	0.007	-0.001	-0.000
	(0.021)	(0.021)	(0.021)	(0.020)
APC exposure x OSRD rate x 1949-1959	0.006	0.006	-0.000	0.001
	(0.020)	(0.019)	(0.020)	(0.019)
APC exposure x OSRD rate x 1960-1975	-0.005	-0.003	-0.010	-0.009
	(0.018)	(0.018)	(0.017)	(0.017)
N	1,894,418	1,894,418	1,792,850	1,792,850
Pseudo R^2	0.279	0.283	0.272	0.276
Y mean	0.592	0.592	0.375	0.375
SD of APC exposure	0.0770	0.0770	0.0770	0.0770
SD of OSRD rate	0.0389	0.0389	0.0389	0.0389
Estimator	PPML	PPML	PPML	PPML
Subclass FE	Yes	Yes	Yes	Yes
Filing-year FE	Yes	Yes	Yes	Yes
NBER main x year FE	Yes	No	Yes	No
NBER subcategory x year FE	No	Yes	No	Yes
ln(universe) x year FE	Yes	Yes	Yes	Yes

Notes: Reference period is 1930-1938. APC exposure is the vested share of pre-April-1942 patents in the subclass; the OSRD rate is the OSRD-funded share of the subclass's 1941-1948 patents (Gross and Sampat). Both are standardized across subclasses, so coefficients are per one-standard-deviation increase. The OSRD rate is set to zero for subclasses with no patents filed in 1941-1948. Outcomes are patent counts, estimated by PPML at the USPTO class-subclass-year level; standard errors are clustered by USPTO class. All columns include subclass fixed effects and ln(universe) by filing-year interactions. Columns (1) and (3) add NBER main-category by year fixed effects; columns (2) and (4) use NBER subcategory by year fixed effects. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

A Supplementary Descriptive Evidence

A.1 Exposure Characteristics

A.1.1 Filing-Year Distribution of Vested Patents

Figure A1 plots the annual count of matched vested patents by filing year. The distribution spans 1914–1942 but is concentrated in the interwar period, with a median filing year of 1931, a mean of 1931.4, and an interquartile range of 1927–1936. The late-period decline should be interpreted cautiously because applications filed near the catalog date had less time to be granted and enter the granted-patent sample, and pending applications are not included in the matched baseline sample.

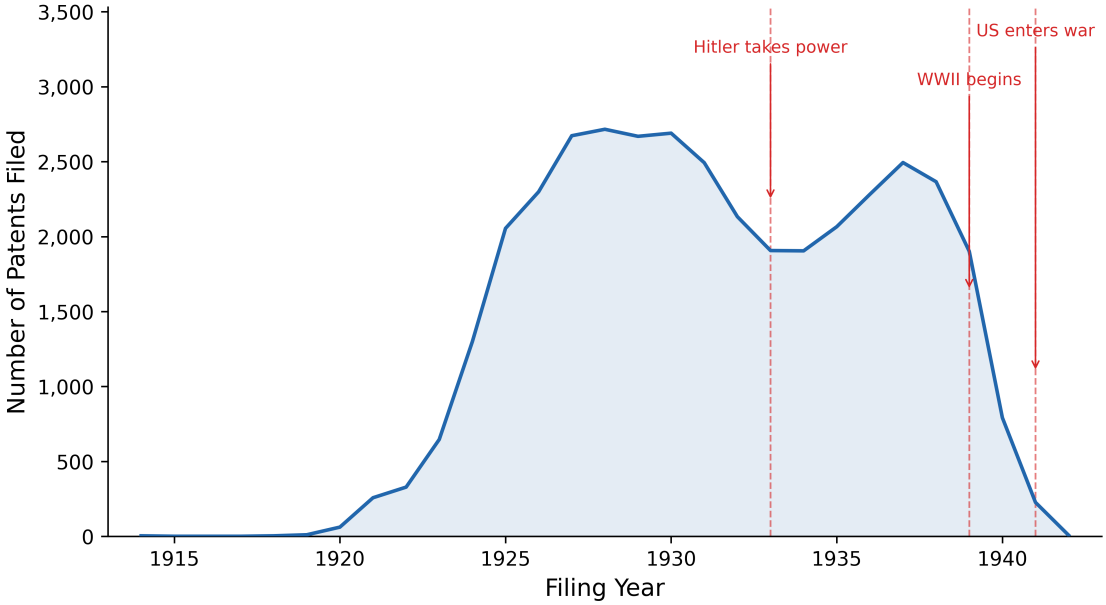


Figure A1: Annual count of matched vested patents by filing year. Dashed lines mark key wartime dates. The data span 1914–1942, and the median filing year is 1931. Counts near the end of the period may also reflect grant lags, the catalog cutoff, and the exclusion of pending applications from the matched baseline sample.

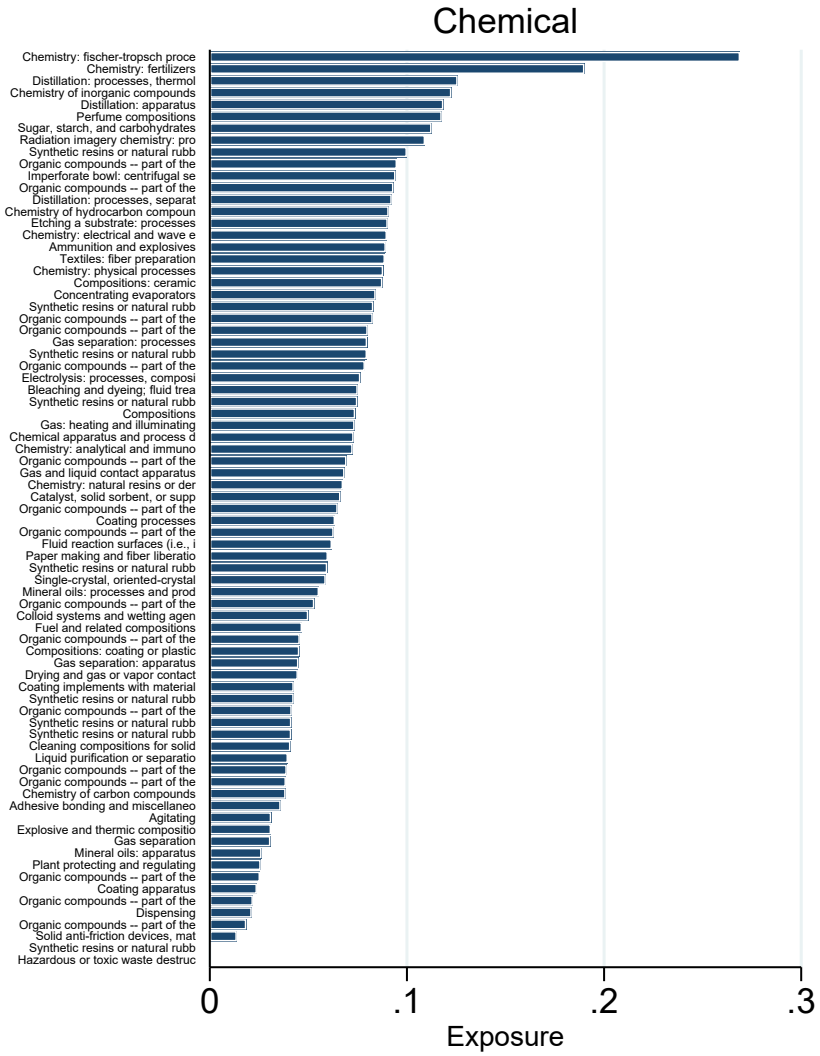


Figure A2: USPTO class-level APC firm exposure: Chemical

Notes: Bars report APC firm exposure at the USPTO class level. Exposure is defined as vested pre-shock firm patents divided by the pre-shock firm-patent universe in each class.

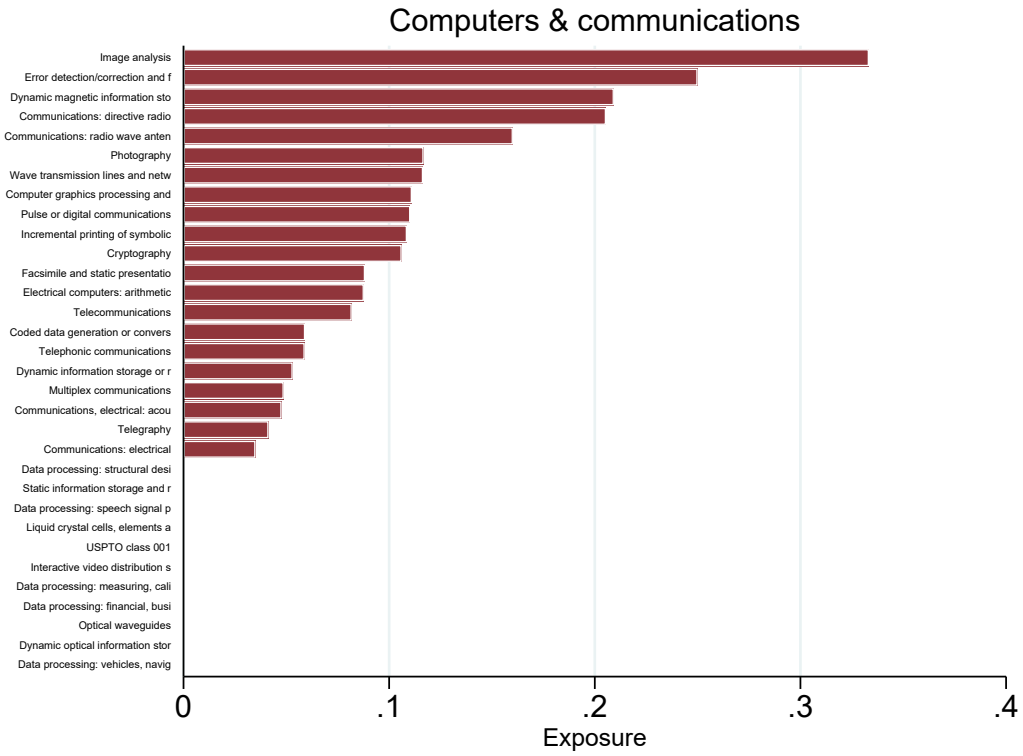


Figure A3: USPTO class-level APC firm exposure: Computers and communications

Notes: Bars report APC firm exposure at the USPTO class level. Exposure is defined as vested pre-shock firm patents divided by the pre-shock firm-patent universe in each class.

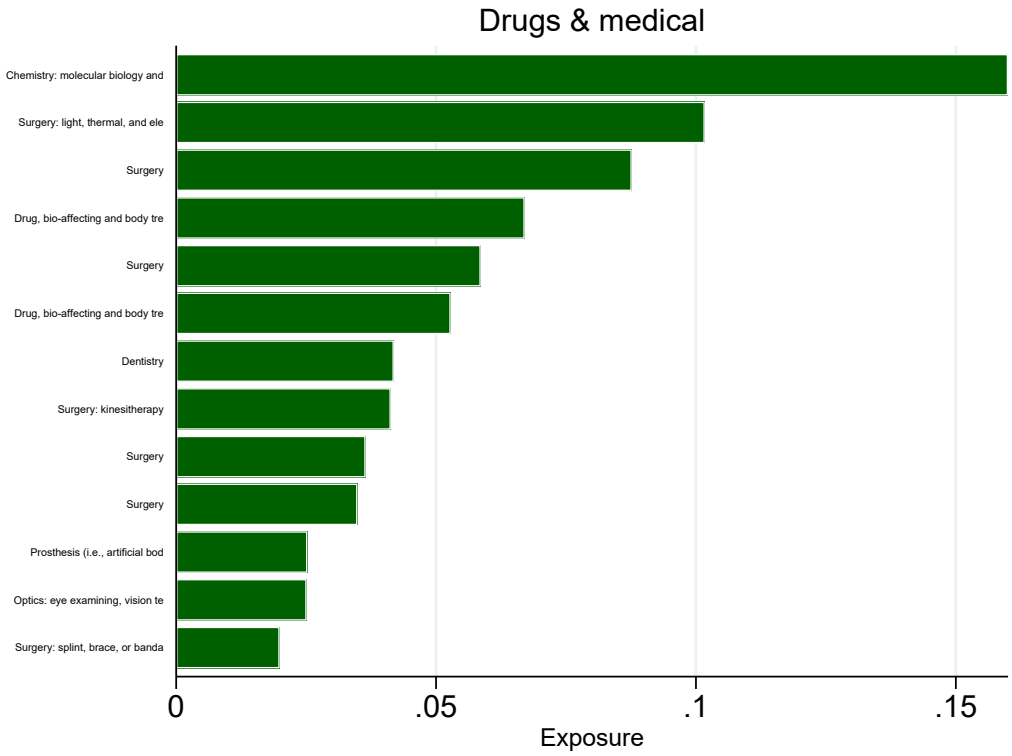


Figure A4: USPTO class-level APC firm exposure: Drugs and medical

Notes: Bars report APC firm exposure at the USPTO class level. Exposure is defined as vested pre-shock firm patents divided by the pre-shock firm-patent universe in each class.

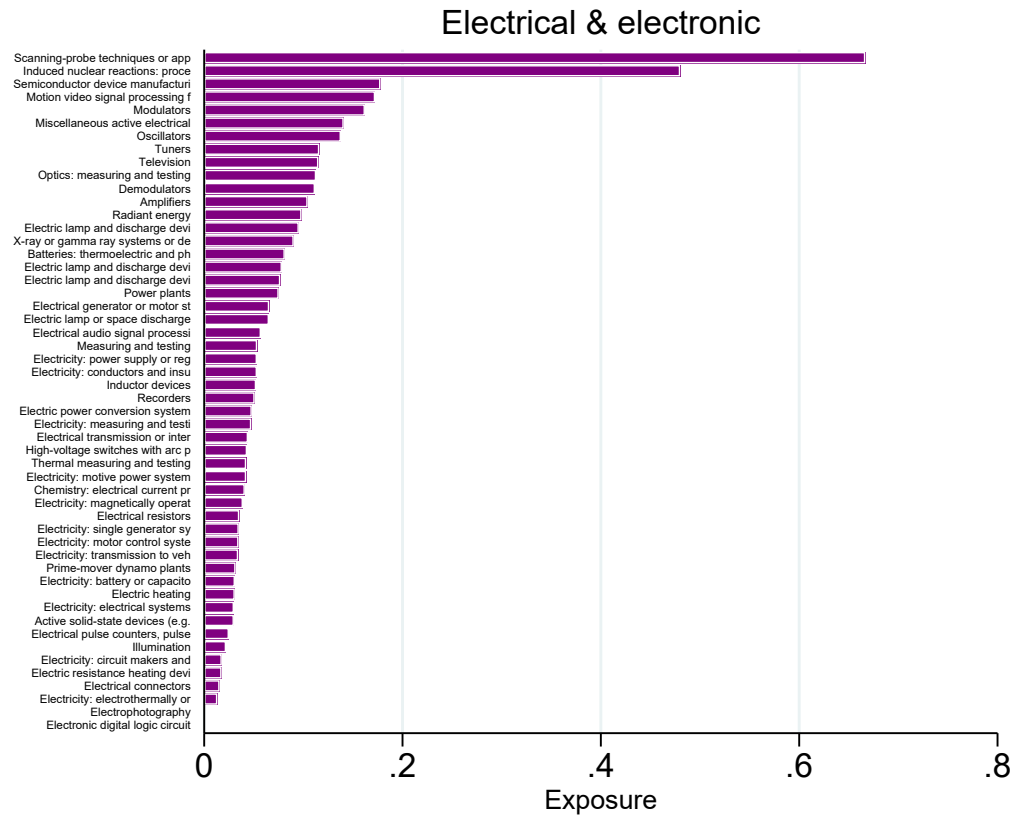


Figure A5: USPTO class-level APC firm exposure: Electrical and electronic

Notes: Bars report APC firm exposure at the USPTO class level. Exposure is defined as vested pre-shock firm patents divided by the pre-shock firm-patent universe in each class.

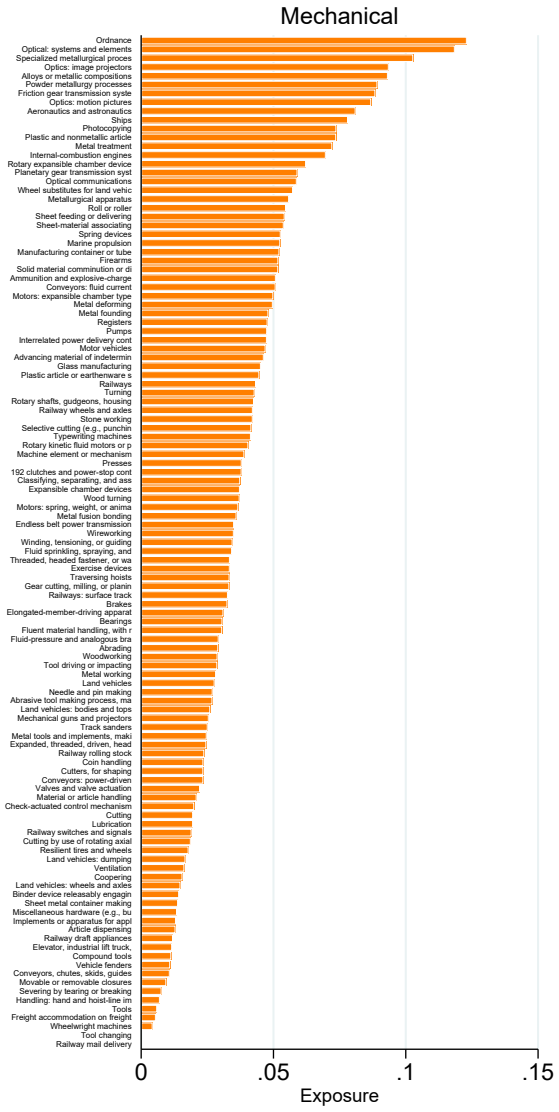


Figure A6: USPTO class-level APC firm exposure: Mechanical

Notes: Bars report APC firm exposure at the USPTO class level. Exposure is defined as vested pre-shock firm patents divided by the pre-shock firm-patent universe in each class.

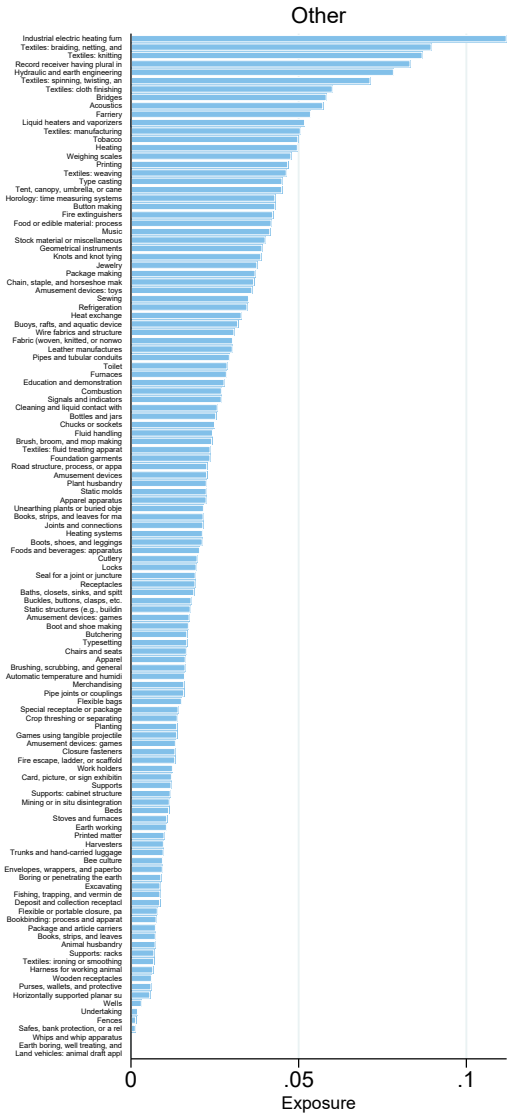
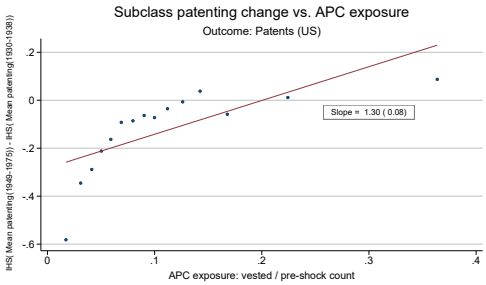


Figure A7: USPTO class-level APC firm exposure: Other

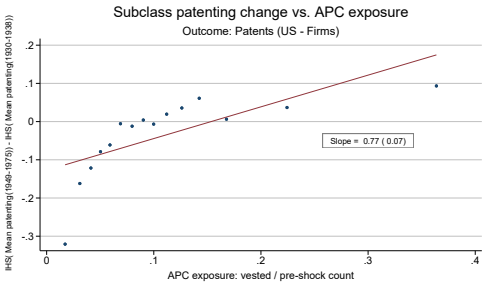
Notes: Bars report APC firm exposure at the USPTO class level. Exposure is defined as vested pre-shock patents divided by the pre-shock patent universe in the corresponding class. Broad NBER categories are constructed from the first digit of the two-digit NBER technology category.

Figure A8: APC exposure and subsequent changes in patenting

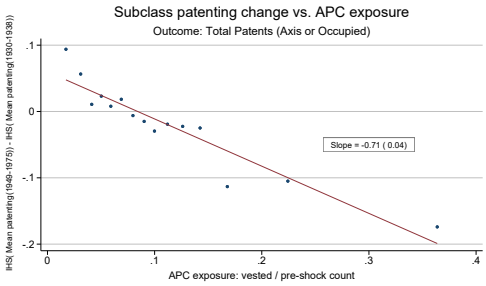
(a) U.S. patents



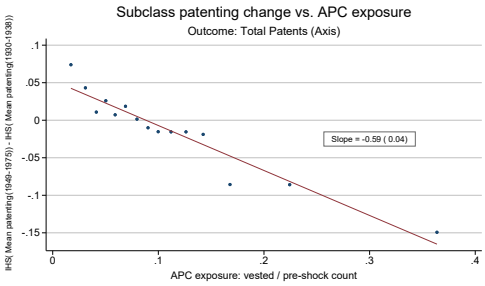
(b) U.S. firm patents



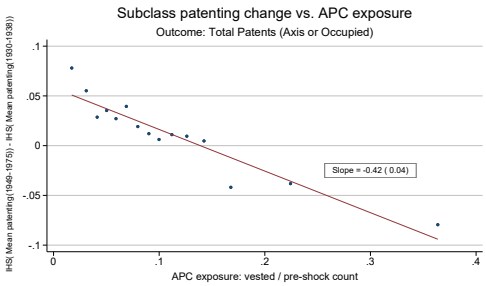
(c) Axis and occupied-origin patents



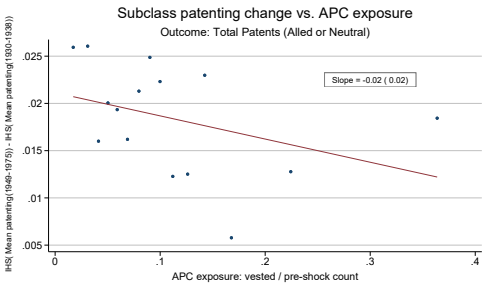
(d) Axis-origin patents



(e) Axis and occupied-origin firm patents



(f) Allied and neutral-origin patents



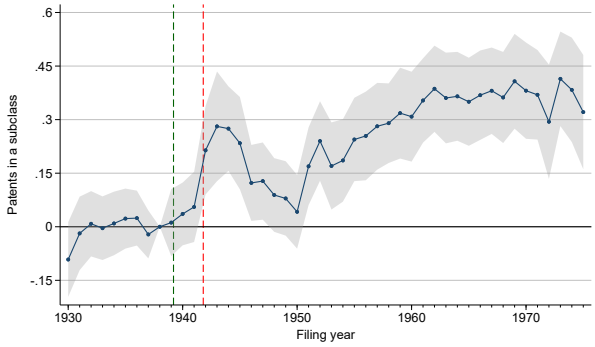
Notes: The unit of observation is a USPTO class–subclass cell. The horizontal axis reports APC exposure, defined as vested patents divided by the pre-shock patent universe. The vertical axis reports the difference between mean inverse-hyperbolic-sine patenting in 1949–1975 and 1930–1938. Points are binned means among subclasses with positive APC exposure. Lines and reported slopes come from subclass-level regressions of the change in patenting on APC exposure; standard errors are reported in parentheses.

B Robustness

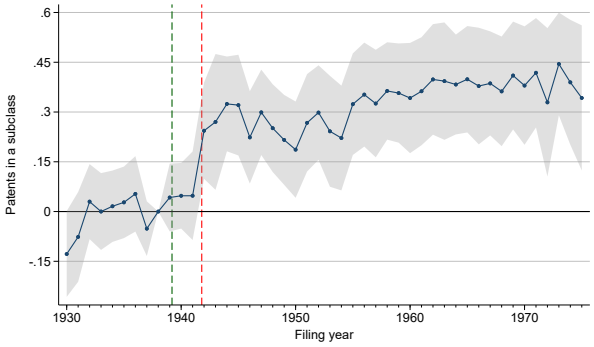
B.1 Citation-Weighted Patent Counts

Figure A9: APC exposure and citation-weighted U.S. patenting

(a) U.S.-origin patents



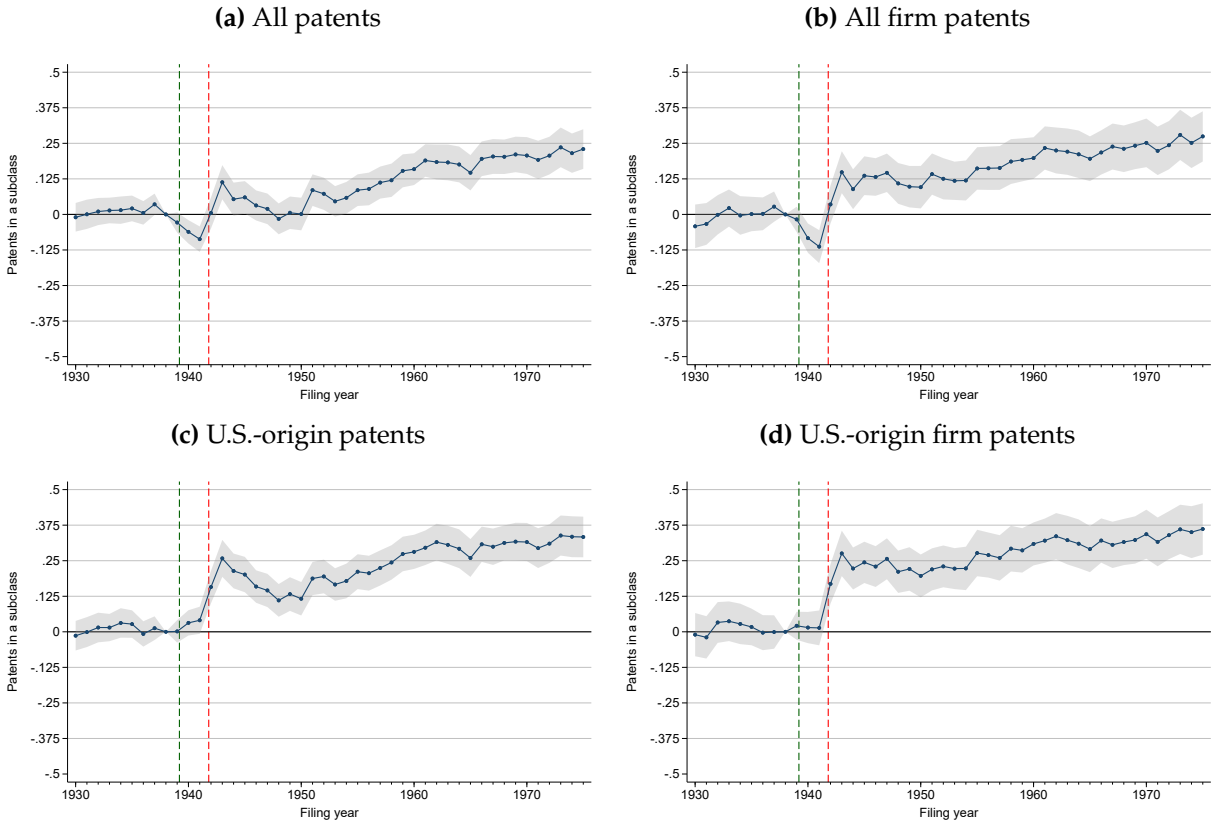
(b) U.S.-origin firm patents



Notes: Outcomes are citation-weighted U.S.-origin patent counts. Estimates are obtained by PPML using any APC exposure as the treatment. The firm outcome counts patents with at least one firm assignee.

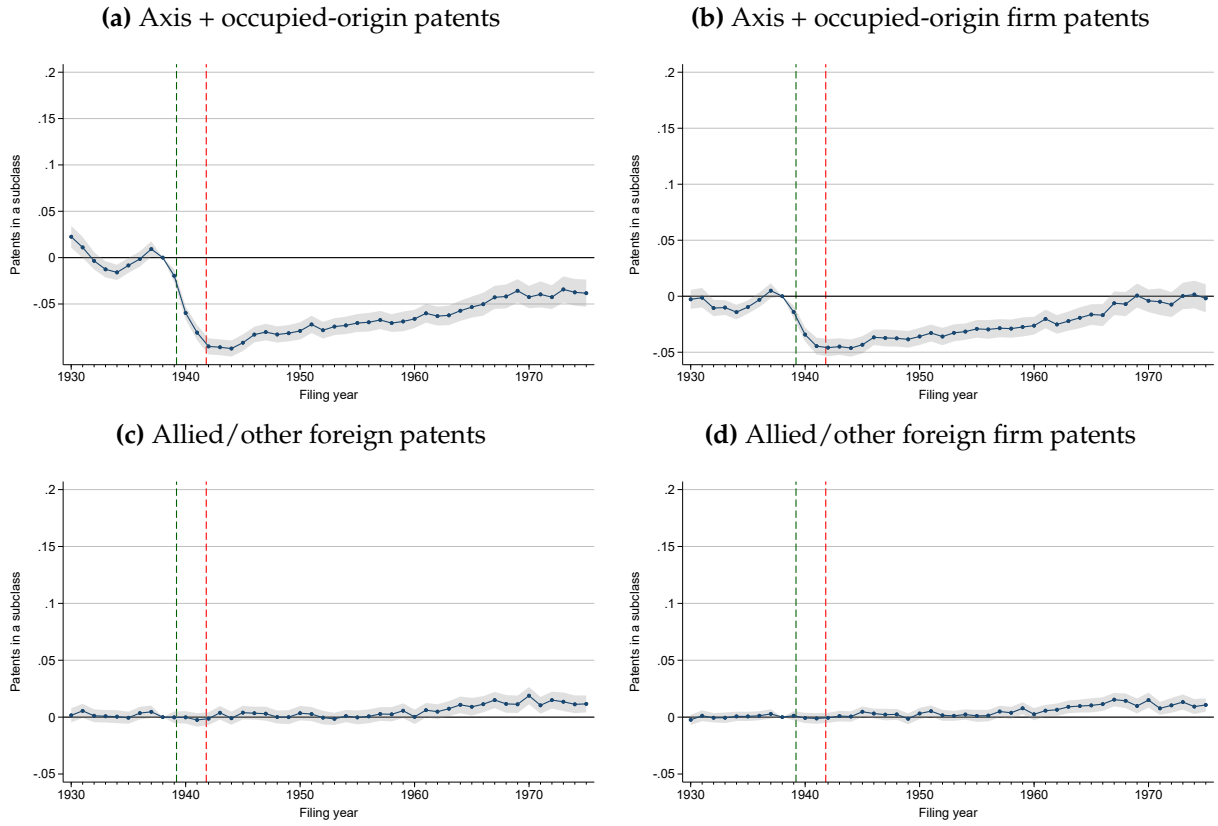
B.2 Controlling for OSRD Exposure

Figure A10: APC exposure and patenting, controlling for OSRD exposure



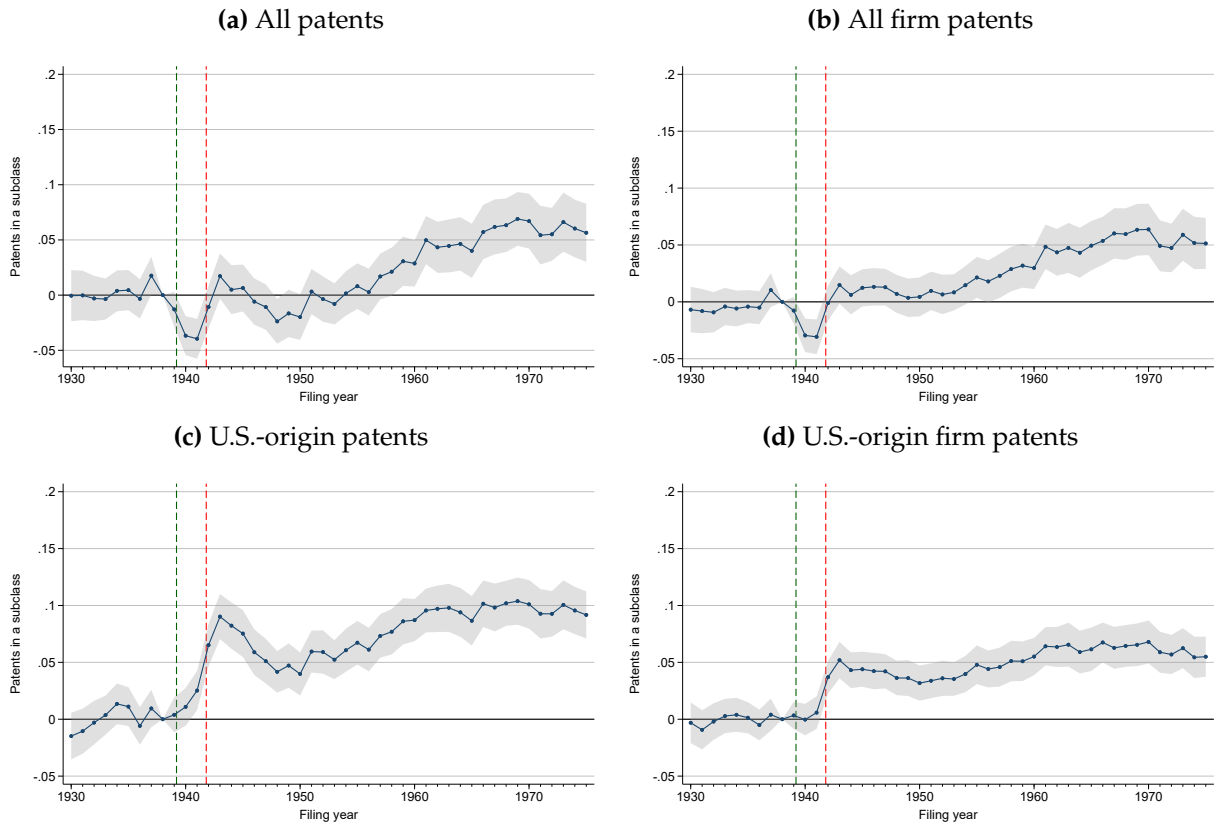
Notes: The panels report subclass-level PPML event studies that add filing-year interactions with the class–subclass OSRD exposure rate. Firm outcomes count patents with at least one firm assignee.

Figure A11: APC exposure and foreign-origin patenting using IHS outcomes



Notes: Outcomes are transformed using the inverse hyperbolic sine. Firm outcomes count patents with at least one firm assignee.

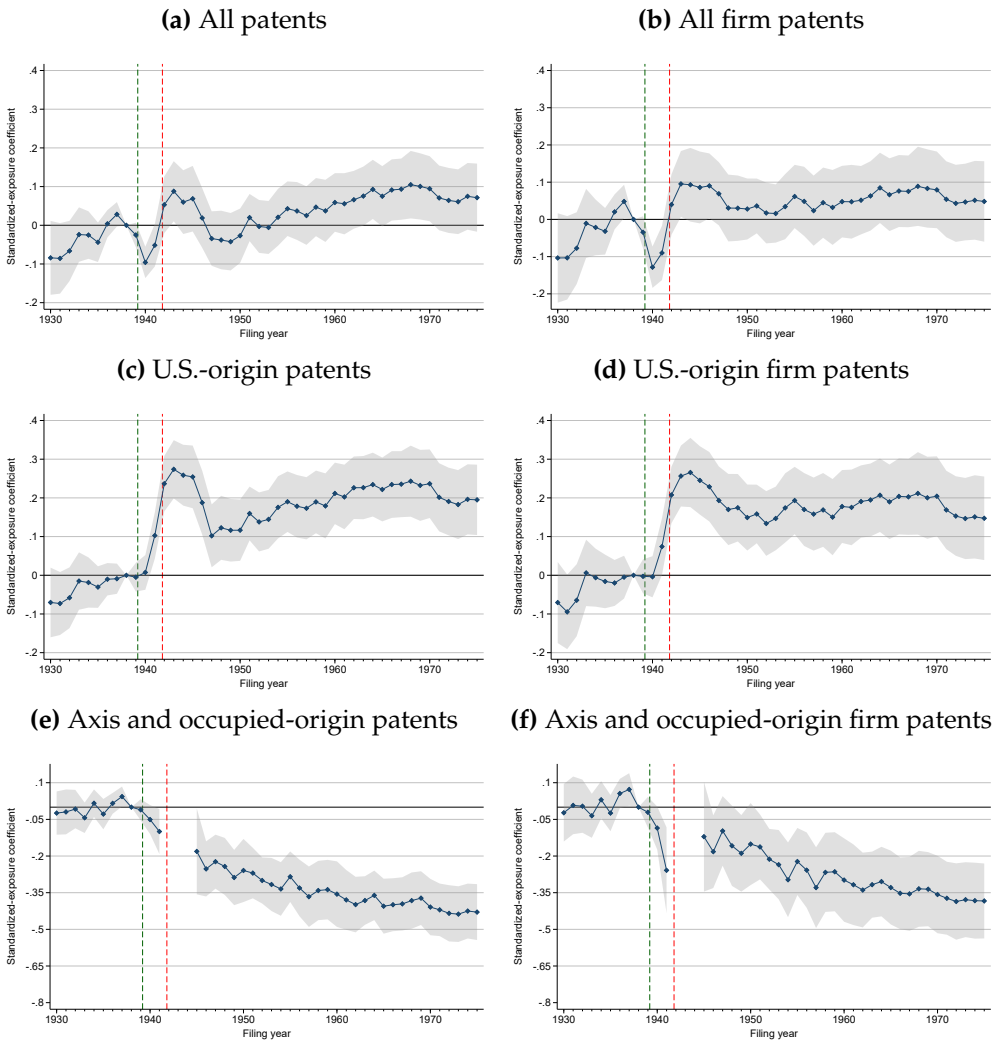
Figure A12: APC exposure and U.S. patenting using IHS outcomes



Notes: Outcomes are transformed using the inverse hyperbolic sine. Panels (a) and (b) include all inventor origins; panels (c) and (d) restrict the outcomes to U.S.-origin patents. Firm outcomes count patents with at least one firm assignee.

B.3 CPC Main-Group Event-Study Evidence

Figure A13: Continuous APC exposure and patenting in CPC main groups



Notes: The unit of observation is a CPC main-group-year cell. The treatment is the standardized continuous APC exposure measure. Firm outcomes count patents with at least one firm assignee.

B.4 Remaining Term of the Vested Patents

Table A1: Subclass PPML: U.S. patenting, continuous APC exposure and vested-patent lifetime

	U.S. patents		U.S. firm patents	
	(1)	(2)	(3)	(4)
APC exposure				
APC exposure x 1939-1941	0.017	0.019*	0.002	0.020
	(0.012)	(0.011)	(0.015)	(0.012)
APC exposure x 1942-1945	0.161***	0.157***	0.158***	0.158***
	(0.018)	(0.017)	(0.020)	(0.020)
APC exposure x 1946-1948	0.120***	0.118***	0.162***	0.159***
	(0.016)	(0.015)	(0.018)	(0.019)
APC exposure x 1949-1959	0.143***	0.141***	0.157***	0.154***
	(0.015)	(0.015)	(0.018)	(0.018)
APC exposure x 1960-1975	0.187***	0.186***	0.184***	0.180***
	(0.016)	(0.016)	(0.019)	(0.019)
Vested-patent lifetime				
Lifetime x 1939-1941	0.051***	-0.040***	0.054***	-0.044***
	(0.013)	(0.012)	(0.017)	(0.017)
Lifetime x 1942-1945	0.029*	0.009	0.032	0.000
	(0.016)	(0.016)	(0.021)	(0.021)
Lifetime x 1946-1948	0.019	0.025	0.017	0.008
	(0.019)	(0.019)	(0.022)	(0.022)
Lifetime x 1949-1959	0.014	0.020	0.002	0.006
	(0.019)	(0.018)	(0.021)	(0.022)
Lifetime x 1960-1975	0.011	0.026	-0.009	0.012
	(0.022)	(0.021)	(0.026)	(0.025)
APC exposure × lifetime				
APC exposure x lifetime x 1939-1941	0.030**	-0.024**	0.045***	-0.020
	(0.012)	(0.012)	(0.016)	(0.015)
APC exposure x lifetime x 1942-1945	0.054***	0.044**	0.052***	0.033*
	(0.017)	(0.017)	(0.020)	(0.020)
APC exposure x lifetime x 1946-1948	0.053**	0.061***	0.047**	0.045**
	(0.022)	(0.021)	(0.023)	(0.023)
APC exposure x lifetime x 1949-1959	0.044*	0.051**	0.041	0.049*
	(0.026)	(0.025)	(0.028)	(0.026)
APC exposure x lifetime x 1960-1975	0.027	0.041*	0.021	0.042*
	(0.022)	(0.022)	(0.024)	(0.023)
N	1,894,418	1,894,418	1,792,850	1,792,850
Pseudo R^2	0.283	0.287	0.276	0.281
Y mean	0.592	0.592	0.375	0.375
SD of APC exposure	0.0770	0.0770	0.0770	0.0770
SD of vested-patent lifetime (years)	4.18	4.18	4.18	4.18
Estimator	PPML	PPML	PPML	PPML
Subclass FE	Yes	Yes	Yes	Yes
Filing-year FE	Yes	Yes	Yes	Yes
NBER subcategory x year FE	Yes	Yes	Yes	Yes
ln(universe) x year FE	Yes	Yes	Yes	Yes
Field age x year FE	No	Yes	No	Yes

Notes: Reference period is 1930-1938. APC exposure is the vested share of pre-April-1942 patents in the subclass, standardized across subclasses. Vested-patent lifetime is the mean remaining lifetime at 4 April 1942 of the subclass's vested patents, with a 17-year term from grant. It is standardized across exposed subclasses and set to its mean in subclasses without vested patents. Field age is the mean filing year of the subclass's patents filed before 4 April 1942, standardized across subclasses. Outcomes are patent counts, estimated by PPML at the USPTO class-subclass-year level; standard errors are clustered by USPTO class. All columns include subclass fixed effects, NBER subcategory by year fixed effects, and ln(universe) by filing-year interactions. Columns (2) and (4) add field age by filing-year interactions. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 Pre-War Concentration, Entry, and Market Structure

The main text shows that APC exposure raised U.S. patenting. This subsection asks whether the response depended on how concentrated a technology field was before the war, and whether it took the form of entry by new firms. We consider two dimensions of concentration. The first is the concentration of all patenting in a field. The second is the concentration of the Axis presence itself, that is, whether the patents of Axis origin in a field belonged to one owner or to many. If vesting mattered because it released technology that a small number of owners had controlled, the response should be larger, and should show up as entry, where that control had been more concentrated.

Unit of analysis. Concentration has to be measured from the distribution of patents across firms within a field and period. USPC class–subclass cells are too small for this purpose: the median subclass records no U.S. firm patent between 1942 and 1948. We therefore use CPC main groups (Section 3.2) and collapse the filing years 1930–1975 into six periods: 1930–1938, which serves as the reference period, 1939–1941, 1942–1945, 1946–1948, 1949–1959, and 1960–1975. The unit of observation is a main group by period. We count firm patent-assignee records. A record is a pairing of a patent with a firm assignee, so a patent with two firm assignees contributes two records, and a patent assigned to several main groups is counted in each, as in the CPC panel. Multiple firm assignees are rare, with 1.002 records per firm patent, so records and patents nearly coincide. A firm is identified by its standardized assignee name, and a patent's origin is the country of its inventor. We refer to patents originating in Axis or enemy-occupied countries as Axis-origin. Pre-war quantities use patents filed before April 30, 1942, the date used to construct exposure for CPC main groups.

Outcomes. We study four outcomes. The first is the number of distinct U.S. firms with at least one U.S.-origin patent in the main group during the period. It rises when firms that had not patented in the field begin to do so, and is our measure of entry. The second is the number of U.S.-origin firm patent-assignee records, which measures the volume of U.S. firm patenting. The third is the concentration of U.S. firm patenting. Let s_f denote firm f's share of the U.S.-origin firm records in a main group and period, let $H = \sum_f s_f^2$ be the Herfindahl index, and let N be the number of U.S. firms. We use the normalized index

$$H^* = \frac{H - 1/N}{1 - 1/N}. \tag{4}$$

The normalization matters because H falls mechanically as the number of firms rises: N firms of equal size give $H = 1/N$. The normalized index equals zero when firms hold equal shares and one when a single firm holds every record, so it isolates the inequality of shares from the number of firms. We set $H^* = 1$ when a single firm holds at least two records and leave it undefined when a main group has fewer than two records in a period. The fourth outcome follows the pre-war incumbent. We identify the firm that held the most firm records in the main group before April 30, 1942, and compute its share of all firm records in the main group, from every country of origin, in each period.

Measures of pre-war concentration. The first measure, the leader share, is the share of a main group's pre-war firm records held by its largest firm. Its mean is 0.214 and its standard deviation is 0.147. It is closely related to the normalized Herfindahl index of pre-war records, with a correlation of 0.91, and is only weakly related to the size of the field, with a correlation of -0.16 with the log number of records. The leader share describes the structure of the field as a whole and not the position of enemy firms. The leading firms were overwhelmingly American: the leading firm held any vested patent in only 124 of the 4,414 main groups in the sample.

The second measure describes the Axis presence directly. The Axis HHI is the Herfindahl index over the owners of a main group's Axis-origin pre-war firm records, with those records as the denominator. It is high when one owner held most of the Axis-origin patents in a field and low when they were spread over many owners. We use Axis-origin records as the denominator by design. The alternative, the share of the whole field held by the largest Axis owner, rises with the size of the Axis presence and is therefore closely related to exposure itself. The Axis HHI measures how concentrated the Axis presence was and not how large it was.

Two features of the patent data require adjustment before the index is computed. First, a patent's origin is the country of its inventor and not the nationality of its owner. About one fifth of Axis-origin records belonged to owners whose patents were mostly of other origin. The largest of these are General Electric, RCA, and Westinghouse, which owned patents on German inventions under their inter-war agreements with German firms. These patents were American property and were not vested. We therefore restrict the set of owners to those for whom at least half of pre-war records are Axis-origin. We also exclude the Chemical Foundation, the American body that held the German patents seized during the First World War. Second, a single enterprise often appears under several assignee names. IG Farben appears under its own name, under the names of its American affiliates, among them General Aniline Works, General Aniline and Film, Agfa Ansco, and Winthrop

Chemical, and under the names of the firms that merged to form it in 1925. Telefunken also appears under its legal name, Gesellschaft für drahtlose Telegraphie. We consolidate fourteen such corporate families, which also include Siemens, Krupp, Zeiss, Schering, Philips, and Bosch, before computing the index.

The index also requires two safeguards. It equals one by construction when a main group has a single Axis owner, which is the typical case when Axis-origin records are few. We therefore keep main groups with at least five Axis-origin records of the retained owners. Of the 3,010 main groups with any such record, 1,297 meet this condition, and 48 of these have a single owner. In addition, the index falls mechanically as the number of records rises. Every specification that uses the Axis HHI therefore includes the log number of Axis-origin records interacted with period, so that the index is compared across main groups with an Axis presence of similar size. In this sample the Axis HHI has a mean of 0.352, a standard deviation of 0.231, and a correlation with APC exposure of 0.04.

Specification. Let j index CPC main groups and p index periods. Let E_j denote continuous APC exposure, defined as in equation (1) for main groups, and let C_j denote one of the two concentration measures. Both are standardized within the estimation sample. We estimate by PPML

$$\mathbb{E}\big[y_{jp} \mid \cdot\big] = \exp\left[\alpha_j + \delta_{g(j)p} + \gamma_p \ln N_j^0 + \sum_{\pi \neq 0} \big(\beta_\pi E_j + \theta_\pi C_j + \phi_\pi E_j C_j\big)\mathbf{1}\{p = \pi\}\right], \quad (5)$$

where α_j are main-group fixed effects, $\delta_{g(j)p}$ are NBER subcategory-by-period fixed effects, N_j^0 is the pre-APC patent universe of the main group, and $\pi = 0$ denotes 1930–1938. Because E_j and C_j do not vary over time, their levels are absorbed by α_j and only their interactions with period are identified. Because both are centered, β_π is the effect of a one-standard-deviation increase in exposure in a main group with average concentration, and ϕ_π is the change in that effect per standard deviation of concentration. The effect of exposure at k standard deviations from mean concentration is $\beta_\pi + k\phi_\pi$. The coefficients θ_π absorb any path that concentrated fields followed regardless of exposure. When C_j is the Axis HHI, the specification adds $\mu_p \ln A_j$, where A_j is the number of Axis-origin records. The estimation sample keeps main groups with at least seven pre-war patents and at least ten pre-war firm records. We use PPML for all outcomes. For the two share outcomes it is a model of the conditional mean of a bounded variable, and the coefficients are proportional changes in that mean. Standard errors are clustered by CPC subclass. For each outcome we also report a specification without the concentration terms.

Results: concentration of the field. Table A2 reports the estimates that use the leader share. Exposure drew additional U.S. firms into exposed fields. In column (5), a one-standard-deviation increase in exposure raises the number of U.S. firms patenting in a main group by 0.150 log points in 1942–1945, or 16 percent, and by about 0.11 log points in each later period. Column (6) shows that during the war this response did not depend on the leader share. After 1949 it did. The interaction is 0.053 in 1949–1959 and 0.045 in 1960–1975. In 1949–1959 the implied effect of exposure on the number of firms is 0.164 (s.e. 0.025) in a main group one standard deviation above the mean leader share and 0.059 (s.e. 0.032) in a main group one standard deviation below it. The entry associated with exposure therefore lasted longer in fields that one firm had led.

Column (2) reports the concentration of U.S. firm patenting. The leader-share rows are large and negative in every period. Fields that were concentrated before the war became less concentrated afterwards whether or not they were exposed, which is the reversion one expects when concentration is measured in one period and followed in the next. The interaction rows measure the additional change associated with exposure. They are negative in every period after 1941 and statistically significant through 1959, at −0.106 in 1942–1945 and −0.084 in 1949–1959. Exposure thus lowered concentration more, or raised it less, the more concentrated a field had been. The sign of the total effect depends on where a field started. In 1942–1945 exposure raised concentration in fields one standard deviation below the mean leader share, by 0.170 (s.e. 0.059), had a statistically insignificant effect in fields one standard deviation above it, and lowered concentration in fields two standard deviations above it, by 0.148 (s.e. 0.047). By 1949–1959 the effect one standard deviation above the mean is −0.086 (s.e. 0.030). Exposure compressed the distribution of concentration across fields.

Column (4) follows the pre-war leader itself. The interaction is negative in every period, between −0.024 and −0.050, and statistically insignificant throughout. We find no evidence that exposure reduced the share of the specific firm that had led a field. This is consistent with the identity of the leaders. Most were American firms whose patents were not vested, so that additional firms patented alongside them.

Results: concentration of the Axis presence. Table A3 reports the estimates that use the Axis HHI. In a main group with an average Axis HHI, exposure has a modest effect on the volume of U.S. firm patenting, 0.133 in 1946–1948, and a statistically insignificant effect on the number of U.S. firms in every period. The response is concentrated in fields where one owner had held the Axis position. The interaction in column (6) is 0.092 in 1942–1945 and lies between 0.080 and 0.095 in each later period. In a main group one standard deviation

above the mean Axis HHI, a one-standard-deviation increase in exposure raises the number of U.S. firms by 0.110 (s.e. 0.043) in 1942–1945, or 12 percent, and the number of U.S. firm patent records by 0.239 (s.e. 0.080), or 27 percent. One standard deviation below the mean, both effects are statistically insignificant. The difference persists: one standard deviation above the mean, the effect on the number of firms is 0.108 (s.e. 0.040) in 1949–1959 and 0.087 (s.e. 0.043) in 1960–1975. The estimates for the concentration of U.S. firm patenting in column (4) change sign across periods, and we draw no conclusion from them.

The two adjustments to ownership matter for magnitudes and not for the conclusion. With the unadjusted index, which treats every assignee name as a separate owner and retains American owners of Axis-origin patents, the interaction for the number of U.S. firms in 1942–1945 is 0.056 (s.e. 0.021). Each adjustment raises the estimate, to 0.066 with consolidated families and to 0.076 with the restriction on owners. This is the pattern expected if mismeasured ownership attenuates the estimate.

The timing deserves comment. The interaction in column (6) is already 0.056 in 1939–1941, a period that begins with the outbreak of war in Europe and precedes vesting. It does not reflect early growth in concentrated fields. One standard deviation above the mean Axis HHI, the effect of exposure on the number of firms is 0.014 (s.e. 0.028) in 1939–1941 and appears only in 1942–1945. The interaction in 1939–1941 arises at the other end of the distribution. One standard deviation below the mean, the number of U.S. firms in exposed fields falls by 0.099 (s.e. 0.034) relative to 1930–1938.

Interpretation. The two tables give a consistent account. Exposure drew additional U.S. firms into exposed fields. Where a single firm had led a field, this entry lasted longer, and the concentration of U.S. patenting fell relative to unexposed fields that had been equally concentrated. Where the Axis position in a field had belonged to one owner, exposure raised both the number of U.S. firms and the volume of their patenting, and where Axis patents had been spread across many owners it did neither. This is the pattern one would expect if vesting made available, through a single licensor, a body of technology that one foreign owner had controlled. Tables 3 and 4 document owners and fields of this kind.

Three qualifications apply. First, the results concern entry and not displacement. The share of the pre-war leading firm does not fall with exposure. Second, the estimates describe heterogeneity in the response and do not isolate a causal effect of concentration. Concentration was not assigned at random, and fields with a concentrated Axis presence were on a different path regardless of exposure: in column (2) of Table A3 the Axis HHI alone is associated with higher U.S. firm patenting, by 0.107 in 1939–1941 and 0.105 in

1946–1948. The interaction is identified under the assumption that this differential path did not itself vary with exposure. Third, the Axis HHI sample is restricted to main groups with at least five Axis-origin records. These are a selected set of fields with a substantial Axis presence, and the estimates should not be extrapolated to fields where that presence was marginal.

Table A2: APC exposure, pre-war concentration, and market structure in CPC main groups

	U.S. firm concentration		Pre-war leader's share		Number of U.S. firms	
	(1)	(2)	(3)	(4)	(5)	(6)
APC exposure						
APC exposure x 1939-1941	0.065*	0.069*	-0.008	0.012	0.060***	0.061***
	(0.035)	(0.039)	(0.028)	(0.029)	(0.014)	(0.014)
APC exposure x 1942-1945	0.007	0.064	-0.014	0.023	0.150***	0.153***
	(0.037)	(0.040)	(0.032)	(0.034)	(0.019)	(0.019)
APC exposure x 1946-1948	0.014	0.041	0.026	0.057	0.114***	0.116***
	(0.037)	(0.039)	(0.039)	(0.040)	(0.018)	(0.018)
APC exposure x 1949-1959	-0.045	-0.002	-0.063	-0.025	0.112***	0.112***
	(0.030)	(0.031)	(0.041)	(0.041)	(0.020)	(0.020)
APC exposure x 1960-1975	-0.045	-0.032	-0.026	0.025	0.113***	0.114***
	(0.028)	(0.028)	(0.047)	(0.041)	(0.023)	(0.023)
Leader share						
Leader share x 1939-1941		-0.102***		0.009		0.034***
		(0.025)		(0.020)		(0.011)
Leader share x 1942-1945		-0.239***		0.122***		0.045***
		(0.028)		(0.024)		(0.016)
Leader share x 1946-1948		-0.281***		0.158***		0.037**
		(0.046)		(0.029)		(0.019)
Leader share x 1949-1959		-0.278***		0.118***		-0.006
		(0.035)		(0.031)		(0.022)
Leader share x 1960-1975		-0.368***		0.179***		-0.002
		(0.038)		(0.048)		(0.029)
APC exposure × leader share						
APC exposure x leader share x 1939-1941		-0.008		-0.024		-0.006
		(0.022)		(0.022)		(0.013)
APC exposure x leader share x 1942-1945		-0.106***		-0.038		0.003
		(0.027)		(0.024)		(0.019)
APC exposure x leader share x 1946-1948		-0.062**		-0.028		0.029
		(0.027)		(0.029)		(0.019)
APC exposure x leader share x 1949-1959		-0.084***		-0.038		0.053***
		(0.023)		(0.025)		(0.020)
APC exposure x leader share x 1960-1975		-0.042		-0.050		0.045**
		(0.027)		(0.032)		(0.023)
Group-periods	23,930	23,930	23,003	23,003	26,472	26,472
Main groups	4,344	4,344	3,934	3,934	4,412	4,412
Pseudo R^2	0.175	0.178	0.198	0.199	0.841	0.841
Y mean	0.071	0.071	0.128	0.128	23.048	23.048
SD of APC exposure	0.0568	0.0568	0.0568	0.0568	0.0568	0.0568
Mean leader share	0.214	0.214	0.214	0.214	0.214	0.214
SD of leader share	0.147	0.147	0.147	0.147	0.147	0.147
Leader-share interactions	No	Yes	No	Yes	No	Yes

Notes: PPML estimates. The unit of observation is a CPC main group by period; the reference period is 1930-1938. APC exposure is the vested share of the main group's patents filed before 30 April 1942, standardized across main groups. Leader share is the share of the main group's firm patent-assignee records filed before 30 April 1942 that were held by its largest firm, standardized across main groups (mean 0.214, standard deviation 0.147). Because both variables are centered, the APC exposure rows in the even-numbered columns give the effect of exposure at the mean leader share, and the interaction rows give the change in that effect per standard deviation of the leader share. Columns (1)-(2): normalized Herfindahl index, (H - 1/N)/(1 - 1/N), of U.S.-origin firm patent-assignee records in the period, where N is the number of U.S. firms; equal to one when a single firm holds at least two records and undefined otherwise when N is below two. Columns (3)-(4): share of the main group's firm patent-assignee records in the period, all countries of origin, held by the firm that was its largest before 30 April 1942. Columns (5)-(6): number of distinct U.S. firms with a patent in the main group in the period. The sample keeps main groups with at least 7 patents and at least 10 firm patent-assignee records filed before 30 April 1942. All columns include main-group fixed effects, NBER subcategory by period fixed effects, and the log of the pre-APC patent universe interacted with period. Standard errors, clustered by CPC subclass, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Concentration of the Axis presence and entry of U.S. firms, CPC main groups

	U.S. firm patent records		U.S. firm concentration		Number of U.S. firms	
	(1)	(2)	(3)	(4)	(5)	(6)
APC exposure						
APC exposure x 1939-1941	-0.056	-0.044	0.083	0.015	-0.039	-0.043*
	(0.053)	(0.050)	(0.066)	(0.072)	(0.026)	(0.025)
APC exposure x 1942-1945	0.109	0.105	0.120*	0.122	0.036	0.019
	(0.071)	(0.071)	(0.072)	(0.078)	(0.036)	(0.036)
APC exposure x 1946-1948	0.127**	0.133**	0.092	-0.003	0.022	0.019
	(0.063)	(0.062)	(0.072)	(0.071)	(0.038)	(0.036)
APC exposure x 1949-1959	0.069	0.067	-0.046	-0.042	0.036	0.019
	(0.068)	(0.067)	(0.055)	(0.056)	(0.040)	(0.040)
APC exposure x 1960-1975	0.067	0.049	-0.100*	-0.102	0.014	-0.009
	(0.072)	(0.070)	(0.058)	(0.069)	(0.045)	(0.046)
Axis HHI						
Axis HHI x 1939-1941		0.107***		-0.013		0.022
		(0.036)		(0.061)		(0.023)
Axis HHI x 1942-1945		0.099		-0.028		0.018
		(0.062)		(0.043)		(0.030)
Axis HHI x 1946-1948		0.105**		0.033		0.062**
		(0.043)		(0.060)		(0.025)
Axis HHI x 1949-1959		0.077*		-0.005		0.019
		(0.043)		(0.045)		(0.027)
Axis HHI x 1960-1975		0.034		-0.053		0.000
		(0.047)		(0.051)		(0.031)
APC exposure × Axis HHI						
APC exposure x Axis HHI x 1939-1941		0.104**		0.088		0.056***
		(0.041)		(0.060)		(0.019)
APC exposure x Axis HHI x 1942-1945		0.134**		-0.042		0.092***
		(0.057)		(0.050)		(0.029)
APC exposure x Axis HHI x 1946-1948		0.117**		0.095**		0.080***
		(0.054)		(0.038)		(0.028)
APC exposure x Axis HHI x 1949-1959		0.121**		-0.011		0.089***
		(0.051)		(0.037)		(0.027)
APC exposure x Axis HHI x 1960-1975		0.134***		-0.026		0.095***
		(0.050)		(0.036)		(0.028)
Group-periods	7,782	7,782	7,552	7,552	7,782	7,782
Main groups	1,297	1,297	1,294	1,294	1,297	1,297
Pseudo R^2	0.906	0.907	0.143	0.144	0.855	0.855
Y mean	110.036	110.036	0.078	0.078	36.524	36.524
SD of APC exposure	0.0688	0.0688	0.0688	0.0688	0.0688	0.0688
Mean Axis HHI	0.352	0.352	0.352	0.352	0.352	0.352
SD of Axis HHI	0.231	0.231	0.231	0.231	0.231	0.231
ln(Axis records) x period	Yes	Yes	Yes	Yes	Yes	Yes
Axis-HHI interactions	No	Yes	No	Yes	No	Yes

Notes: PPML estimates. The unit of observation is a CPC main group by period; the reference period is 1930-1938. APC exposure is the vested share of the main group's patents filed before 30 April 1942, standardized across the main groups in this sample. Axis HHI is the Herfindahl index over the owners of the main group's Axis and occupied-country-origin firm patent-assignee records filed before 30 April 1942, with those records as the denominator; it is standardized across the main groups in this sample (mean 0.352, standard deviation 0.231). Owners are consolidated into corporate families, and the set of owners is restricted to those whose pre-1942 records are at least half of Axis or occupied-country origin; see the text. The exposure and interaction rows are read as in Table A2. Columns (1)-(2): number of U.S.-origin firm patent-assignee records in the period. Columns (3)-(4): normalized Herfindahl index of those records, as in that table. Columns (5)-(6): number of distinct U.S. firms with a patent in the main group in the period. The sample keeps main groups with at least 7 patents, at least 10 firm patent-assignee records, and at least 5 Axis or occupied-country-origin records of the retained owners, all filed before 30 April 1942. All columns include main-group fixed effects, NBER subcategory by period fixed effects, the log of the pre-APC patent universe interacted with period, and the log number of Axis records interacted with period. Standard errors, clustered by CPC subclass, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.